

The Rollover Clock: Debt Maturity, the Central-Bank Balance Sheet, and the Debt Limit of a Reserve-Currency Sovereign

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<https://github.com/dingningpei/rollover-clock> (Appendix D).

Abstract

When can a sovereign that borrows in its own reserve currency keep rolling its debt over, and what stops it if it cannot? We separate two layers of the answer. In the **fiscal layer**, debt is locally stable if and only if the primary surplus offsets at least a share $\phi^* = 1 - g/(r + \psi b)$ of the marginal interest cost of debt; we prove that this threshold does not depend on maturity, for any repricing kernel. In the **inflation layer**, surprise inflation substitutes for a missing fiscal response. How much it takes depends on how fast the consolidated liabilities of the Treasury and the central bank reprice, which we call the *rollover clock*, and on how much of them inflation can erode, including currency.

We measure the clock security by security for the United States from 1980 to 2025. It predicts the Treasury's average interest rate within 0.05–0.07 percentage points, reproduces the 2022–25 rise in interest costs and the Federal Reserve's operating loss, and shows that in 2021–25 markets priced far less inflation into new debt than full Fisher repricing implies.

Today's fiscal threshold ($\phi^* \approx 0.47$) is back in its pre-2008 range. What has changed is the fiscal response, down from about 0.39 before 2004 to about zero, and the funding of the Fed: QE replaced currency, which inflation erodes, with interest-bearing reserves, and accounts for 50–80% of the rise since 2007 in the inflation needed per unit of relief. Covering the fiscal gap after a permanent 1pp rate rise would take 1.0–1.1 pp of extra inflation a year for a decade, the most since at least 1980; holding debt/GDP stable at CBO's baseline deficits, about 4 pp.

The same measures for the United Kingdom and Japan show that the inflation layer is set by the consolidated balance sheet, not by debt management. Despite the longest gilts of the three, the UK's consolidated ratio matched that of the United States at the 2021 QE peak, and Japan's 2024 move to paying interest on reserves more than doubled its ratio. The binding constraint is not default but the cost of the inflation tax.

JEL: E52, E58, E62, E63, H62, H63

Keywords: rollover risk; debt maturity; consolidated government balance sheet; quantitative easing; fiscal limits; inflation tax; debt sustainability

1. Introduction

Can an issuer that replaces maturing high-coupon debt with new low-coupon debt keep doing so forever? For a sovereign that issues the world's reserve currency and whose central bank can buy its bonds, default is a choice rather than a constraint. The question is then not whether a debt limit exists but what form it takes. This paper argues that the limit has two layers, and that they depend on the debt's maturity in opposite ways.

The fiscal layer is maturity-free. In a transparent model of consolidated debt dynamics, the market rate rises with debt and the average rate on outstanding debt reprices toward it at a speed set by maturity. Debt is then locally stable if and only if

$$(1 - \phi)(r + \psi b) < g,$$

where ϕ is the share of marginal interest cost offset by the primary surplus and $r + \psi b$ is the marginal interest cost of debt. With $\phi = 0$ this is the condition of Mian, Straub and Sufi (2025). We show that the repricing speed cancels from it for any maturity structure. Shorter debt makes a rate shock bite faster but does not change whether the system is stable. What it does change is timing. When the primary surplus has a ceiling, maturity decides how many years pass before a large rate shock breaks the limit (Section 3.5).

The inflation layer depends on maturity and on what the government owes. If the fiscal response falls short of the threshold, the gap must be closed some other way, and for a reserve-currency sovereign that other way is inflation. Sustained surprise inflation erodes only nominal liabilities that have not yet repriced. Debt that rolls over at new, Fisher-adjusted rates escapes; currency, which never reprices, does not. The inflation needed to replace a missing fiscal offset therefore depends on the rollover clock $P(h)$, the share of a permanent rate shock that has reached the average rate after h years, and on the erodible share $E(h)$ of the balance sheet:

$$\Delta\pi^{req}(H) = (\phi^* - \hat{\phi})^+ \Delta r \frac{\int_0^H P(h) dh}{\int_0^H E(h) dh}.$$

Two implications follow. First, inflation on debt only buys time: debt reprices, so its erosion runs out as the horizon lengthens, and only currency can be taxed permanently. At end-2025, covering the fiscal gap with a permanent inflation tax on currency would take about 5.6 pp of extra inflation a year, before any flight from cash. Second, the relevant balance sheet is the consolidated one. A central bank that holds Treasuries against currency turns interest-bearing debt into a zero-interest liability that inflation erodes. One that holds them against interest-bearing reserves does the opposite: it shortens the maturity of the government's liabilities to the private sector and takes them out of the inflation-tax base. The inflation substitute then becomes costlier, even while Treasury-only maturity statistics lengthen.

Measurement. We build the clock from the Monthly Statement of the Public Debt security by security for 2001–2025, net the Fed's holdings CUSIP by CUSIP, add reserves and reverse repos as overnight liabilities, and add currency to the base that inflation erodes. Hand-transcribed maturity distributions from the Treasury Bulletin extend the series to 1980. The reconstruction matches official totals exactly in every year, and the Fed's holdings match its balance sheet exactly from 2007.

Validation. Three tests check the clock. In a backtest over 22 year-end origins from 2001 to 2024, projecting the average interest rate on marketable debt from the portfolio known at the origin and realized yields gives a mean absolute error of 0.047 pp at 12 months, against 0.279 pp for a clock based only on weighted-average maturity. Frozen at end-2021, the clock tracks the 1.9 pp rise in the Treasury's average rate over four years within about 0.05 pp, and overnight repricing of reserves against the frozen Fed book reproduces the Fed's cumulative loss: a deferred asset of –\$245bn predicted against –\$243bn actual. Finally, from the end-2020 balance sheet, the 2021–23 inflation surprise transferred about 11% of GDP from holders of nominal liabilities by 2023. That is 1.8 times what full Fisher repricing implies, because new debt was priced for 2–3% inflation while prices rose 6–7% a year; higher real rates then took most of it back, leaving a net transfer of 4% of GDP by 2025.

Findings. On a common 1980–2025 timeline, the fiscal threshold today is ordinary. At about 0.47 in 2023–25 it is within its 1980–2007 range of roughly 0.1–0.56; the exception was 2008–21, when the threshold was negative. The fiscal response is not ordinary. Before 2004 the legislated offset of higher interest costs was about 0.39, roughly enough to meet the threshold, and the 1990 and 1993 budget agreements belong to that regime. Since 2004 it has been about zero (Auerbach and Yagan 2024). Our own Bohn-type regressions confirm a break at 2004 in direction but cannot pin down its size, so we also report every result under a single fiscal response common to all years.

The third finding concerns the Fed. Before 2008, consolidating with the Fed roughly halved the inflation needed per unit of relief, because the Fed's Treasuries were financed by currency. Since 2009 the consolidated and Treasury-only requirements have been about equal. From end-2007 to end-2025 the consolidated ratio rose by half, and QE's reserve funding accounts for 50–80% of that rise, depending on the order of decomposition.

Together these make the inflation needed to cover the fiscal gap after a permanent 1pp rate rise the highest in the sample in 2023–25: 1.01–1.12 pp a year for ten years, against 0.61 in 2007 and 0.23 in 1981. The ranking is driven by the inflation layer, not by the post-2004 collapse of the fiscal response: with one common response in every year it holds for any value below about 0.35, and it holds in 99% of draws when the parameters are drawn from their plausible ranges. In levels, holding debt/GDP stable at CBO's baseline primary deficits and today's rates would take about 4 pp of extra inflation a year for a decade, against about 3 pp in 2019 and nothing in 2007. Each

point of sustained surprise inflation erodes about \$100bn a year, so a primary shortfall of about \$400bn a year is far beyond what a modest inflation overshoot can cover.

Counterfactuals rank the levers. Restoring the pre-2004 fiscal response cuts the required inflation by about 84%. Among balance-sheet policies, ending interest on reserves has the largest effect, a third, because reserves rejoin the zero-interest base; but it amounts to a tax on banks and would change how the Fed sets rates. Undoing QE, so that the Fed holds only as many Treasuries as it has currency outstanding, cuts the requirement by 18%, and a Fed shift to bills cuts it by 12% because it leaves longer debt in private hands. Raising the Treasury's bill share to 30% raises it by 15%, and losing 50–100bp of convenience yield by 8–14%.

Three sovereigns. Section 6 builds the same measures for the United Kingdom (2007–2025, gilt by gilt, net of the Bank of England's Asset Purchase Facility) and Japan (fiscal years 2020–2024, JGB by issue, net of the Bank of Japan). On their own debt the three governments differ widely: the UK's gilts average 14–16 years and a quarter are index-linked, and its own-debt inflation ratio is about 0.9 against 2.1–2.7 for the United States. Consolidation erases most of the difference. At the QE peak the consolidated ratios were 2.47 in the UK, 2.46 in the United States and 1.99 in Japan. How the central bank funds its bonds is decisive. Japan's zero-rate reserve tier kept its ratio near 0.8 until the March 2024 reform, which raised it to about 2 with almost no change in the government's own debt. The fiscal threshold is where the three differ: the UK is the only one with r above g , and Japan's threshold depends almost entirely on how rates respond to debt. The UK's 2021–25 inflation repeats the U.S. result: new debt was priced for far less inflation than followed, so the surprise transferred 1.8 times what full Fisher repricing implies, and the extra interest on reserves then took back most of the transfer.

Contribution. None of the ingredients is new on its own. Debt dynamics with endogenous rates come from Mian, Straub and Sufi (2025) and Lorenzoni and Werning (2019); consolidated Treasury–Fed maturity from Greenwood, Hanson, Rudolph and Summers (2014), the TBAC (2020, 2026) and the OBR (2021); maturity and the inflation tax from Cochrane (2001, 2022), Hilscher, Raviv and Reis (2022) and Barro and Bianchi (2026); and fiscal responses to debt service from Bohn (1998), Auerbach and Yagan (2024) and Eichengreen, Menuet and Donnat (2026). The paper puts them together. It proves that the fiscal threshold is maturity-free and derives a closed-form inflation requirement in which maturity enters only through the measured clock. It builds a long series, 1980–2025, of the full consolidated repricing profile for the United States, generalizing the scalar reset measures used in debt management and testing it out of sample; we found no earlier series of this kind. And it gives a year-by-year, two-layer distance to the limit that separates what fiscal policy must do from what balance-sheet policy changes, quantifying the effect of consolidation on the inflation requirement, which Barro and Bianchi (2026) note would be desirable but for which "data are not available."

Section 2 reviews related work and Section 3 presents the model. Section 4 builds and tests the clock. Section 5 presents the two-layer limit from 1980 to 2025, our own estimates of the fiscal response, the level of inflation that would hold debt stable today, and the time to the limit when the surplus has a ceiling. Section 6 builds the same measures for the United Kingdom and Japan. Section 7 reports counterfactuals and policy options, and Section 8 concludes.

2. Related literature

Debt limits with endogenous rates. Mian, Straub and Sufi (2025) derive fiscal space when $R - G$ rises with debt. Their free-lunch condition, $R < G - \phi$ with ϕ the sensitivity of $R - G$ to log debt (ψb in our notation), is the $\phi = 0$ case of ours; their maturity extension works through the convenience yield of long versus short debt in steady state, not through repricing speed. Lorenzoni and Werning (2019) show that responsive fiscal rules and longer maturity can rule out slow-moving debt crises, through default equilibria. Li and Merkel (2026) show that QE can shift a default boundary inward by depleting central-bank capital; their §4.8 notes that in transition longer maturity reduces rollover pressure, since only a fraction $1/n$ of the stock comes due each period, while in steady state it shrinks the fiscal dividend from convenience yields. Our setting has no default. Maturity leaves the stability threshold unchanged and instead governs the inflation alternative. In the tradition of Sargent and Wallace (1981), Davig, Leeper and Walker (2011) and Leeper and Walker (2011), the non-default limit is where fiscal adjustment stops and inflation must take over, and we make that boundary measurable. Related work on sustainability and debt capacity includes Blanchard (2019), Mehrotra and Sergeyev (2021), Reis (2021) and Ghosh et al. (2013), and the convenience-yield literature (Krishnamurthy and Vissing-Jorgensen 2012; Choi, Kirpalani and Perez 2026; Jiang et al. 2024).

Consolidated maturity and QE. Greenwood et al. (2014) measure consolidated Treasury–Fed duration and show that Treasury's maturity extension offset about a third of QE's reduction in privately held duration (35% from a 2007 baseline; 63% from December 2008). The Treasury Borrowing Advisory Committee (2020) computed consolidated maturity and duration, and since 2022 Treasury's refunding materials report a consolidated weighted-average next rate reset, which TBAC (2026) reviews; it treats the part of SOMA that funds reserves, ON RRP and other interest-bearing Fed liabilities as repricing overnight. The OBR (2021) documents the same shortening for the UK, and CBO (2022), Levin, Lu and Nelson (2022), Cavallo et al. (2019) and d'Avernas et al. (2024) discuss the fiscal risk it creates. Del Negro and Sims (2015) and Hall and Reis (2015) ask when a central bank's balance sheet needs fiscal support, and Bassetto and Messer (2013) trace the fiscal consequences of paying interest on reserves; our consolidated clock shows those consequences in the erosion base, since remunerated reserves took the Fed's book out of the inflation-tax base in 2008. In July 2022, Fed staff projected the deferred asset to peak at about \$60 billion, with a tail of about \$180 billion (Anderson et al. 2022). We generalize the scalar reset

measures to the full repricing profile, build it from 1980, and test it on realized outcomes, including the Fed's 2022–25 losses.

Maturity and the inflation tax. In the fiscal theory of the price level, maturity determines whether fiscal news shows up as inflation now or later (Cochrane 2001, 2022, 2023), and monetary and fiscal policy jointly determine the price level (Sims 2013). Barro and Bianchi (2026) scale fiscal spending shocks by debt times duration in explaining 2020–23 inflation. Hilscher, Raviv and Reis (2022) and Aizenman and Marion (2011) show that short U.S. maturity limits how far inflation can erode debt. Reis (2017a) argues that QE, by bringing more debt due, lowers the price increase a fiscal crisis requires, and Reis (2017b) sets out how far a central bank can relieve fiscal burdens, including through seigniorage on the currency base. Krause and Moyen (2016) show in a New Keynesian model with a maturity structure that a higher inflation target lowers real debt substantially only if the change is close to permanent; our inflation layer measures the same channel on the actual consolidated balance sheet, with currency in the erosion base. Section 3.4 reconciles the signs in this literature: a one-time level jump is maturity-neutral, a transitory surprise favors short debt, and sustained inflation favors long debt. We invert the erosion calculation into a required inflation rate as a function of the measured clock and quantify how consolidation shifts it.

Optimal maturity. Angeletos (2002), Bhandari, Evans, Golosov and Sargent (2017) and Faraglia, Marcet, Oikonomou and Scott (2019) study how the maturity structure should be chosen to insure the budget. We take the structure as given and measure its consequences.

Fiscal responses. Bohn (1998, 2008) estimates a positive U.S. response of surpluses to debt; his coefficient maps exactly into ours (Section 5.1). Eichengreen, Menuet and Donnat (2026) find that surpluses respond to debt service more than to debt stocks, and more strongly when $r > g$; their object is a log-log elasticity, and they describe their evidence as descriptive. Auerbach and Yagan (2024) find that legislated U.S. fiscal feedback, including the response to lagged net interest, was substantial before 2004 and has been about zero since. We use their estimates as the baseline ϕ , check them with our own regressions in Section 5.3, and show which conclusions hold without them.

3. Model

3.1 Setup

All variables are ratios to GDP. The government is consolidated: Treasury plus central bank. b is net debt to the private sector (marketable Treasuries held outside the central bank, plus reserves and reverse repos), $\bar{r}$ is the average effective rate on b , r the market rate on new issuance, s the primary surplus, and g growth.

$$\dot{b} = (\bar{r} - g)b - s, \quad r = r^* + \psi(b - b^*), \quad s = s^* + \phi(\bar{r}b - r^*b^*).$$

Repricing. If debt matures at rate λ and grows through net issuance at rate $\dot{B}/B$, the average rate obeys, exactly,

$$\dot{r} = (\lambda + \dot{B}/B)(r - \bar{r}).$$

We verify this against a cohort-level simulation (maximum error 4×10^{-7}). Near a steady state the repricing speed is $\Lambda = \lambda + g$: new debt issued to keep pace with growth is priced at the new rate. For a general maturity structure with repricing distribution $F(h)$, the share of a permanent rate shock that reaches the average rate after h years is

$$P(h) = 1 - (1 - F(h))e^{-gh}.$$

We call $P(h)$ the *rollover clock*.

3.2 The fiscal layer is maturity-free

Linearizing around the steady state gives

$$\det J = \Lambda [g - (1 - \phi)(r^* + \psi b^*)],$$

and the steady state is stable if and only if $(1 - \phi)(r^* + \psi b^*) < g$. Λ cancels.

The result holds for any repricing kernel. The characteristic equation is $x = a + c\psi K(x)$, with $a = (1 - \phi)r^* - g$, $c = (1 - \phi)b^*$, and K the Laplace transform of the kernel. Since $|K(x)| \leq 1$ for $\text{Re } x \geq 0$, and the condition implies $|a| > |c\psi|$, no root lies in the closed right half-plane.

When fiscal adjustment instead responds with a lag, a numerical scan over 3,180 parameter combinations with $\lambda \in [0.05, 5]$ found no case in which λ changes stability. Maturity also barely moves the asymptotic divergence rate when the system is unstable.

The minimum stabilizing offset is

$$\phi^* = 1 - \frac{g}{r + \psi b}.$$

What Result 1 does and does not say. CBO (2022), the OBR (2021) and TBAC (2020, 2026) show that QE shortened the maturity of the consolidated government's liabilities and so raised the budget's sensitivity to interest rates. That is correct, and the rollover clock measures it. What does not follow is that QE or shorter maturity moved the threshold at which debt becomes unstable. In this model class maturity is a filter between the market rate and the average rate: it shapes the path of interest costs after a shock but not the slow root that decides stability.

The result has three boundaries.

- If the debt-rate schedule depends on the duration the private sector holds (a term-premium channel), maturity moves r and through it ϕ^* . The effect runs through r , which we measure, not through repricing speed.

- If the fiscal rule responds to projected rather than realized interest, the response is faster but the threshold is the same.
- If the primary surplus has a ceiling, maturity decides when a large shock breaks the limit. Section 3.5 makes this precise.

3.3 The inflation layer

Let a permanent real-rate rise Δr add extra interest $\Delta r \cdot b \cdot P(h)$, where P is the clock of all interest-bearing liabilities; inflation-indexed debt reprices its real coupon at maturity like any other bond.

Sustained surprise inflation $\Delta \pi$ lowers the real value of nominal liabilities that have not yet repriced. Two parts of the consolidated balance sheet qualify:

- non-indexed debt N , with its own clock P_N ;
- currency C , which pays no interest and never reprices. Before October 2008 this also includes reserves.

Inflation-indexed debt cannot be eroded. Per unit of interest-bearing debt b , the erodible share at horizon h is

$$E(h) = \frac{N}{b} \left[1 - P_N(h) \right] + \frac{C}{b}.$$

Replacing an unfinanced share u of the extra interest over horizon H requires

$$\Delta \pi^{req}(H) = u \Delta r \frac{\int_0^H P}{\int_0^H E}.$$

If all debt were nominal and there were no currency, the denominator would be $\int (1 - P)$.

Corollary (inflation on debt only buys time). $\int_0^\infty [1 - P_N] = \int_0^\infty [1 - F_N] e^{-gh} dh$ is finite, so the debt part of the erosion base runs out. Only the currency part grows with H . As $H \rightarrow \infty$,

$$\Delta \pi^{req}(H) \rightarrow u \Delta r \frac{b}{C},$$

the permanent inflation tax on currency that covers the gap. Without currency, $\Delta \pi^{req}(H) \rightarrow \infty$. At end-2025 $b/C \approx 12$, so with $u = 0.46$ the limit is about 5.6 pp per year indefinitely, before any fall in currency demand. In 2007 it was about 1.9.

The currency base is where the reserve-currency status enters the inflation layer directly. About 44% of U.S. currency is held abroad, so close to half of the inflation tax on currency is paid by foreign holders. Counting only domestically held currency would raise the 2025 consolidated ratio from 2.18 to 2.46.

Combined metric. Local stability requires offsetting the share ϕ^* of marginal interest cost. With a historical offset $\phi < \phi^*$, the missing share is $u = (\phi^* - \phi)^+$. This gives the two-layer distance to the

limit:

$$\Delta\pi^{gap}(H) = (\phi^* - \hat{\phi})^+ \Delta r \frac{\int_0^H P}{\int_0^H E}.$$

The approximations are:

- ϕ^* is evaluated at r rather than $r + \Delta r$, which is conservative;
- flows are matched undiscounted;
- Δr is exogenous;
- currency demand does not respond to inflation, which makes the requirement a lower bound;
- H is finite.

3.4 Which inflation? Reconciling the maturity sign

Let $w(s)$ be the share of nominal debt payments due at s and $p(s)$ the cumulative surprise in the price level. Real erosion of the debt is $N \int w(s)p(s)ds$; currency C loses $p(s)$ of its value for as long as the price level stays higher. How maturity matters depends on how persistent the price-level path is:

Price-level path	Erosion (debt + currency)	Effect of shorter maturity
Permanent one-time jump, $p(s) = \Delta p$	$(N + C) \cdot \Delta p$	none
Transitory surprise until τ , then reversed	$N \cdot \Delta p \cdot F_N(\tau)$	helps (Reis 2017a: "more of the debt coming due")
Sustained inflation over H , $p(s) = \Delta\pi \cdot s$	$(N \cdot D_N + C \cdot H) \cdot \Delta\pi$	hurts through the debt term (Cochrane; Barro–Bianchi; HRR; this paper)

Here D_N is the average time to repricing of nominal debt over the horizon. A reversed surprise leaves currency holders whole, while a sustained one taxes currency throughout.

QE therefore makes a front-loaded surprise more effective and a persistent inflation tax less effective. Because our question is whether inflation can stand in for a missing fiscal response over a horizon, we use the sustained object. We also report the one-time level jump, $\Delta p^{\text{req}} = u \cdot \Delta r \cdot \int_0^H P / (N/b + C/b)$, which depends on maturity only through the cost of the shock.

3.5 When maturity matters: a ceiling on the surplus

Governments cannot raise the primary surplus without limit; Ghosh et al. (2013) call the flattening of the response at high debt fiscal fatigue. Let the surplus have a ceiling s_{max} . After a permanent rise Δr in the marginal rate, the primary balance that holds debt/GDP constant is

$$s^{req}(h) = (\bar{r}(h) - g) b - g \cdot z - \Delta\pi b E(h), \quad \bar{r}(h) = \bar{r}_0 + (r + \Delta r - \bar{r}_0) P(h),$$

where z is the zero-interest base relative to GDP, $g \cdot z$ its seigniorage, and $\Delta\pi$ an optional sustained surprise inflation. The limit is broken when $s^{req}(h)$ first exceeds s_max , at the time to the limit $T(\Delta r)$.

Two properties follow.

- *Whether* the limit is broken does not depend on maturity: it happens whenever the long-run requirement $(r + \Delta r - g)b - g \cdot z - \Delta\pi \cdot z$ exceeds s_max .
- *When* it is broken depends only on the rollover clock, because the requirement climbs toward its long-run value at the speed $P(h)$. A faster clock shortens T .

Inflation enters the same way. A sustained surprise lowers the requirement by $\Delta\pi \cdot b \cdot E(h)$, which shrinks as the debt reprices, so it lengthens T but, unless the currency base alone can carry the gap, does not remove the limit. This is the sense in which inflation buys time. Section 5.5 measures T .

4. Measuring the rollover clock

4.1 Data

Treasury, 2001–2025. MSPD Table 3 (marketable), year-end snapshots, one row per CUSIP.

- Repricing dates are maturity for bills, notes, bonds and TIPS, and one week for FRNs.
- The security-level sums match the official unmatured totals for all 25 year-ends, to within 2×10^{-14} %. One MSPD label is misspelled ("Tresasury Floating Rate Notes", 2016); we handle it.

Federal Reserve, 2003–2025.

- SOMA Treasury holdings by CUSIP, including TIPS inflation compensation, from the New York Fed. They match H.4.1 Treasuries held outright exactly from 2007, and within 0.25–0.51% in 2003–06.
- Reserve balances and reverse repurchase agreements from H.4.1. Both reprice overnight.
- Currency pays no interest, so it is not part of interest-bearing debt b . It enters the erosion base of the inflation layer (Section 3.3), measured as H.4.1 currency in circulation in the same week as reserves. Reserves paid no interest before October 2008, so through 2007 they are counted with currency.
- The Treasury General Account is intragovernmental and excluded.
- The Fed's MBS (about \$2tn at end-2025) are funded by reserves. Consolidated debt includes those reserves but does not net the MBS: they are a fixed-rate asset funded by a floating-rate

liability, so the interest-rate exposure is real. Section 7 reports the no-QE counterfactual both with debt/GDP fixed and with the MBS-funding reserves removed.

- About 44% of U.S. currency was held abroad at end-2025 (33% at end-2007), according to the Financial Accounts of the United States. The inflation tax on it falls on foreign holders. We keep all currency in the erosion base, because the transfer accrues to the U.S. government either way.

1980–2002. Treasury Bulletin table FD-5 (FD-7 before 1983): marketable debt held by private investors, which excludes the Fed and government accounts, by remaining-maturity bucket.

- Reserves paid no interest before October 2008, so privately held marketable debt is the consolidated interest-bearing liability for that period.
- We transcribed 24 December rows from FRASER scans. Every row's buckets sum to its total within ± 1 (in \$mn).
- Checks against the exact data:
 - The December 2003 FD-5 total (2,908,029) matches MSPD minus SOMA (2,910,053) within 0.07%.
 - Bucketing understates the 10-year inflation-layer ratio by a stable 0.077 (s.d. 0.016) on 2003–07 exact data. We correct for this.
- FD-5 does not separate inflation-indexed notes (first issued in 1997). We remove them from the erosion base by maturity bucket, using the Treasury Bulletin's FD-2 totals and the maturities of the TIPS issues outstanding at each year-end. Before 2003 this also removes the Fed's small TIPS holdings (8% of TIPS in 2003); scaling the removal by the private share instead changes the ratio by at most 0.006.
- The zero-interest base is currency in circulation plus reserve balances, December averages (FRED CURRCIR and RESBALNS). It was 27–33% of privately held marketable debt in 1980–2002, against 8% in 2025.

Precedent. Security-level accounting of U.S. debt returns and its effect on debt/GDP dynamics goes back to Hall and Sargent (2011); we use the same MSPD-based building blocks to measure repricing speed rather than realized returns.

Other inputs. Yields are H.15 constant-maturity series; the –9999 missing-value code (20-year CMT, 1987–93) is treated as missing. Official average interest rates on marketable debt and the auction record are from FiscalData. GDP is from BEA.

4.2 The clock, 1980–2025

Figure 1 shows $P(1)$, the share of a permanent rate shock that reaches the average rate within a year.

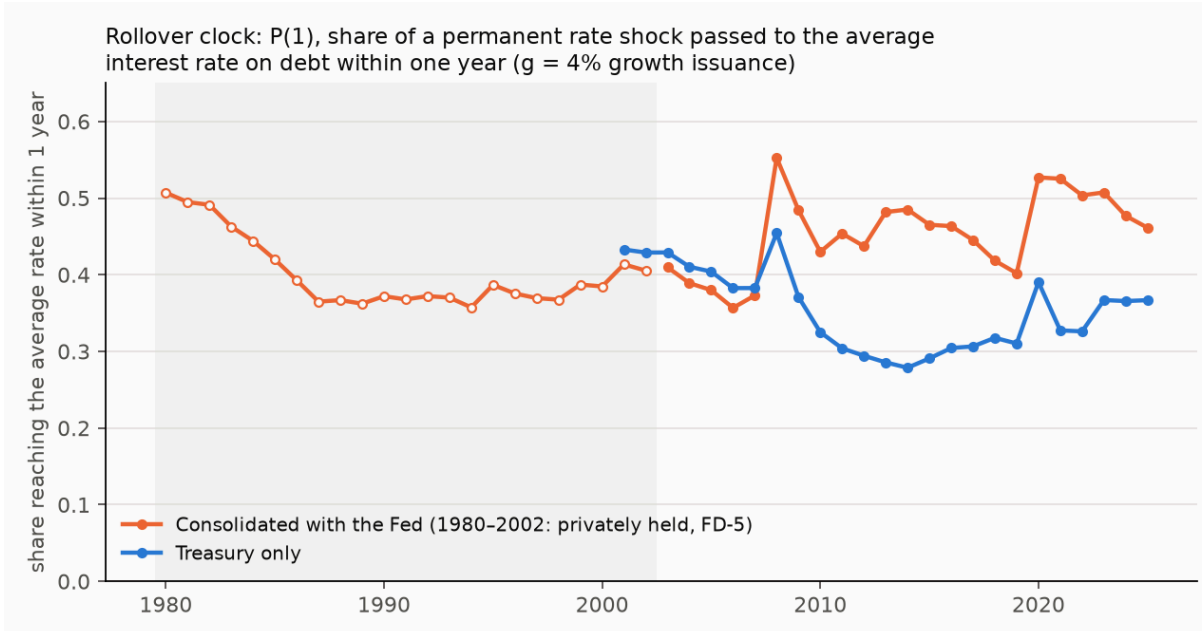


Figure 1. Rollover clock $P(1)$. Orange: consolidated. For 1980–2002 this is privately held debt from FD-5 buckets (open markers); for 2003–2025 it is MSPD net of SOMA plus reserves and RRP. Blue: Treasury only (MSPD, 2001–).

Four features stand out.

1. The clock slowed steadily through the 1980s as Treasury lengthened maturity: $P(1)$ fell from 0.51 in 1980 to about 0.37 by 1987.
2. Before 2008 the consolidated clock was slightly *slower* than the Treasury-only clock, because the Fed held mostly bills.
3. QE reversed this. In 2021 the consolidated $P(1)$ was 0.53 against 0.33 for the Treasury alone, and consolidated WAM was 4.1 years against 6.0. Treasury was lengthening while the Fed was shortening.
4. QT narrowed the gap, but in 2025 the consolidated $P(1)$ was still about 9 points higher.

4.3 Interest-cost pass-through I: historical backtest

From each year-end origin t (2001–2024; 2004 is omitted because the official figure is missing), we project the official average interest rate on marketable debt for 36 months:

- surviving securities keep their rate;
- maturing principal and net new borrowing are reissued at realized H.15 yields, using the auction mix of year t .

We score the change in the rate against two benchmarks: a scalar clock based on weighted-average maturity (WAM), which is the information in maturity summaries, and no change.

Table 1. Backtest: mean absolute error (bias) of the predicted change in the average rate, pp

Horizon	Clock, realized borrowing	Clock, trend borrowing	WAM-only clock	No change	Origins
12 months	0.047 (−0.003)	0.091 (+0.068)	0.279 (−0.108)	0.389 (+0.118)	22
24 months	0.066 (−0.005)	0.155 (+0.116)	0.502 (−0.254)	0.693 (+0.167)	21
36 months	0.071 (−0.019)	0.170 (+0.140)	0.620 (−0.431)	0.815 (+0.146)	20

The full repricing profile cuts the error of a maturity summary by a factor of three to nine.

The largest ex-ante misses come from unanticipated bill-financed deficits: 2007 (+0.52 pp at 12 months) and 2019 (+0.32 pp). They shrink to +0.14 and +0.08 pp once realized borrowing is used.

4.4 Interest-cost pass-through II: the 2022–25 tightening from end-2021

We freeze both balance sheets at end-2021, before the first rate increase, and feed in realized rates and quantities.

Treasury (Figure 2A).

- The clock predicts a 0.94 pp rise in the average rate by December 2022 (actual 0.89) and 1.96 pp by December 2025 (actual 1.93).
- The WAM-only clock predicts 0.17 and 1.57.

Federal Reserve (Figure 2B). The model is as follows.

- The asset book is frozen at end-2021:
 - Treasuries at their coupons;
 - MBS at the end-2021 book coupon of 2.48%;
 - TIPS inflation compensation as realized;
 - premium amortization net of discount accretion.
- Reserves reprice overnight at the IORB rate and reverse repos at the ON RRP rate. Both come from the FOMC target range: IORB is the top of the range minus 10bp; ON RRP is the bottom plus 5bp, and the bottom itself from 19 December 2024.
- Other expenses (operating expenses plus dividends minus other income) are taken year by year from the Reserve Banks' combined financial statements.

The model reproduces the Fed's deferred asset at each year-end:

	2022	2023	2024	2025
Predicted	−18	−142	−226	−245
Actual	−18	−132	−215	−243

(\$bn.) Premium amortization matters: without it, the model understates losses by about 40%.

Component check against the combined financial statements:

- Modeled reverse-repo interest expense matches the statements within about 1% in every year (\$40.8bn vs \$42.0bn in 2022; \$103.3bn vs \$104.3bn in 2023), which validates the policy-rate rule.
- Modeled interest on reserves is 6–10% below the statements' "depository institutions and others" line, which also covers other deposits.
- Modeled interest income on the frozen book is \$6–23bn below actual.

The close deferred-asset fit therefore partly reflects offsetting errors on the income and expense sides. The out-of-sample claim rests mainly on the timing and size of the expense leg.

For comparison, in July 2022 Fed staff projected the deferred asset to peak at about \$60 billion in their baseline and about \$180 billion in the tail (Anderson et al. 2022). The outcome passed \$240 billion. Because the accounting, fed the realized rates, reproduces the outcome, the gap between that projection and the outcome lies in the assumed rate path.

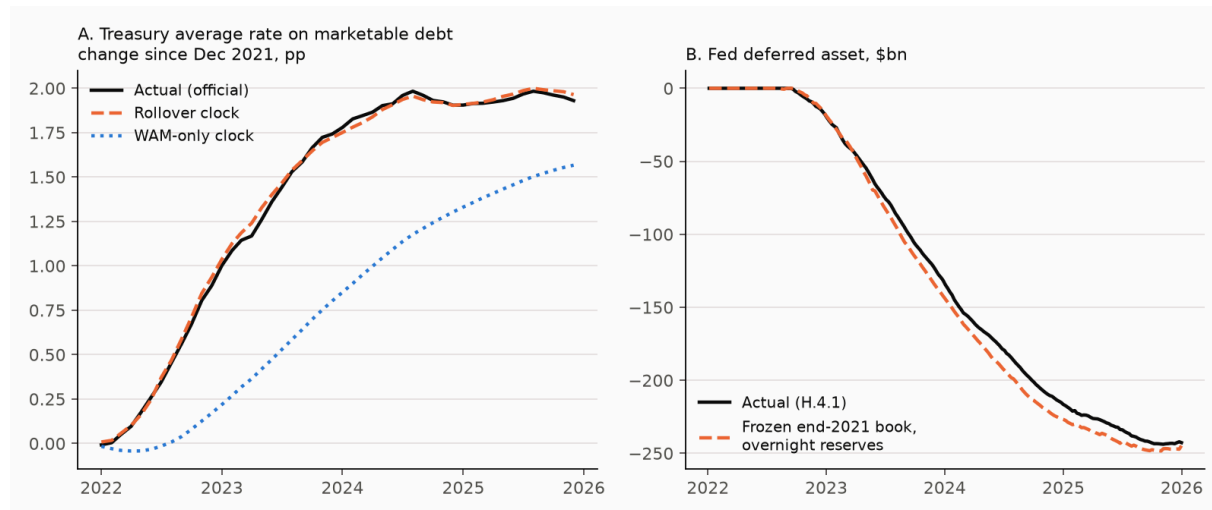


Figure 2. Out-of-sample test from end-2021. A: change in the official average rate on marketable debt; the clock (dashed) against the WAM-only clock (dotted). B: Fed deferred asset; frozen end-2021 asset book with reserves repricing overnight.

The consolidated clock is thus validated where QE matters most. The 2022–25 rate shock reached consolidated interest costs through a Treasury clock that passed through about 0.9 pp within a year,

and through an overnight Fed clock that passed through at once and appeared in the budget as lost remittances.

4.5 The inflation layer in 2021–25

The two tests above check how rate shocks pass through to interest costs. The inflation layer rests on a different assumption: surprise inflation erodes nominal liabilities until they reprice, and repriced debt then pays Fisher-adjusted rates. The inflation surprise of 2021–23 lets us test it.

We freeze the consolidated balance sheet at end-2020. Expected inflation is the end-2020 five-year breakeven, 1.87%, and the surprise is realized CPI inflation minus 1.87 (October 2025, not published because of the federal shutdown, is interpolated). The real transfer from holders of nominal government liabilities to the government, relative to a no-surprise path, accrues each month as the surprise times nominal liabilities, less the extra interest paid when debt reprices:

- nominal liabilities N are privately held non-indexed marketable debt, reserves, reverse repos and currency;
- extra interest is computed with the rollover-clock projection of Section 4.3, under realized borrowing.

Four runs differ only in the yields at which debt reprices:

- **A**, no surprise: the end-2020 forward curve;
- **B**, the model's assumption: forwards plus the surprise realized over each new security's life (full Fisher repricing);
- **D**, inflation as priced: forwards plus the actual change in breakeven inflation, with real yields at their forwards;
- **C**, actual yields.

Table 2. Real transfer to the consolidated government from the 2021–25 inflation surprise
(cumulative, % of 2020 GDP)

End of	Cumulative surprise (pp of price level)	Gross erosion	Model: full Fisher repricing (B)	Model, analytic $s \cdot b \cdot E$	Inflation as priced (D)	Actual (C)
2021	5.1	5.4	3.1	2.7	5.1	5.4
2022	9.4	10.3	5.5	4.7	9.5	9.4
2023	10.8	12.0	6.1	5.2	10.9	8.2
2024	11.7	13.2	6.6	5.5	11.8	6.2
2025	12.5	14.2	6.8	5.7	12.5	4.2

Three findings follow.

1. **The accounting holds up.** Gross erosion of nominal liabilities reached 12% of 2020 GDP by end-2023, about \$2.6tn. The analytic formula $s \cdot b \cdot E(h)$, applied to the end-2020 stock, gives 5.2% by 2023 against 6.1% for its numerical counterpart B; the difference is new borrowing, which the analytic version omits.
2. **Markets repriced far less than the model assumes, so the surprise went further.** Five-year breakeven inflation averaged 2.7% in 2021–22 and peaked at 3.4%, while CPI inflation averaged 6.6%. New debt was therefore issued at rates that did not compensate for the inflation that followed. With inflation compensation as actually priced (D), the transfer reached 10.9% of GDP by 2023, about 1.8 times the 6.1% implied by full Fisher repricing (B). The inflation layer is thus conservative for an inflation that markets do not anticipate. Its object is a sustained inflation that new issues price (Section 3.4), and an unanticipated burst cannot be repeated at will.
3. **Higher real rates took most of it back.** The five-year real yield rose from -1.5% at end-2020 to an average of 1.7% in 2024–25. With actual yields (C), the transfer peaked at 9.4% of GDP in 2022 and fell to 4.2% by end-2025, as debt repriced at higher real rates. This is the rate layer at work: the transfer from the inflation surprise was largely reversed by the real-rate shock that followed it.

5. The two-layer limit, 1980–2025

5.1 Parameters

- **$\psi = 3\text{bp per pp of debt/GDP}$** , range 2–4.5. Sources: Laubach 2009 (3–4); Engen and Hubbard 2005 (≈ 3); Plante, Richter and Zubairy 2025 (3.0–3.5); Bhatt et al. 2026 (3–4); Bi, Phillot and Zubairy 2026 (2.8 at peak, from identified supply shocks). CBO's 2bp (Neveu and Schafer 2024) is the low case.
- **$\hat{\phi}$, the historical fiscal response.** The baseline is 0.39 through 2003 and 0 from 2004. These are Auerbach and Yagan's (2024) estimates of the legislated response to lagged net interest (0.386, s.e. 0.131, for 1984–2003; -0.31 , s.e. 0.41, for 2004–24; their Table 3).
 - The evidence is thinner than the point values suggest. The pre-2004 coefficient comes from semi-annual data and falls to about -0.12 once their projected-surplus control is added, and the post-2004 coefficient cannot reject 0.39.
 - We therefore also report results under a single $\hat{\phi}$ common to all years (0, 0.15, 0.25, 0.35, 0.39), and we estimate the response ourselves in Section 5.3.
 - De Groot, Holm-Hadulla and Leiner-Killinger (2015) imply an offset of about 0.7 over ten years for European countries. We treat that as an upper bound.

- **Mapping from Bohn's coefficient.** Bohn (1998) estimates ρ in $s = \rho b + \text{controls}$. In our model the surplus responds to interest cost with slope ϕ , so at a steady state $ds/db = \phi(r + \psi b)$, and
 - $\phi = \rho / (r + \psi b)$;
 - Bohn's sustainability condition $\rho > (r + \psi b) - g$ is then exactly our condition $(1 - \phi)(r + \psi b) < g$;
 - a Bohn coefficient of 0.02 at a marginal cost of 8% corresponds to $\phi = 0.25$, the intermediate value we use.
- **r** is the steady-state marginal cost of the existing structure: each security's original tenor priced at current yields. Before 2003 we use remaining-maturity pricing, corrected by its measured 0.28 pp gap on 2003–2025.
- **g** , expected nominal growth, is proxied by realized forward ten-year nominal GDP growth through 2015. From 2016 we use the December FOMC Summary of Economic Projections: the longer-run median real GDP growth (1.8–1.9%) plus the 2% inflation objective. Trailing ten-year growth and an ex-ante measure (trailing real growth plus the Cleveland Fed's ten-year expected inflation) are sensitivities (Table 4). In 2013–15, when both measures exist, the SEP-based g (4.0–4.3%) was below realized forward growth (5.2–5.4%), which was inflated by 2021–22.

5.2 Results

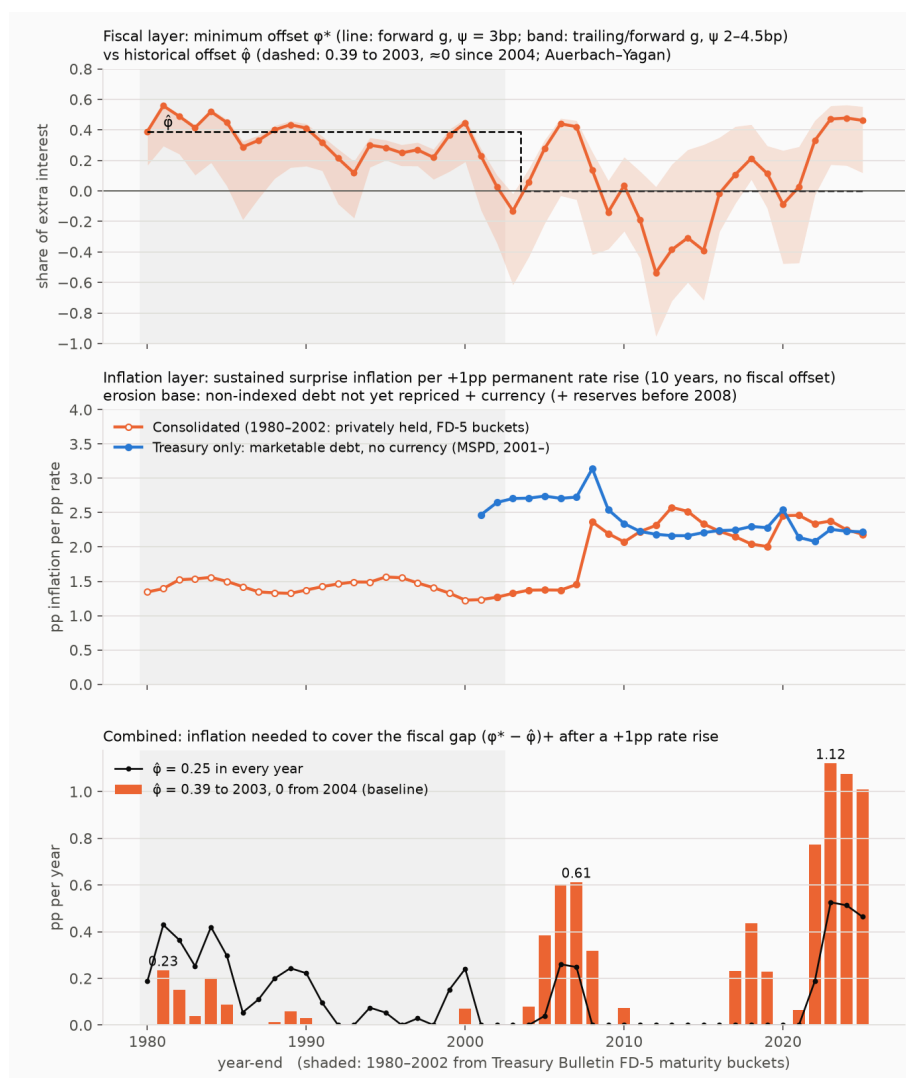


Figure 3. Two-layer limit. Top: fiscal threshold ϕ^* (line; band spans g and ψ) and historical offset $\hat{\phi}$ (dashed). Middle: inflation-layer ratio $[P/\Delta E]$, consolidated (erosion base includes currency, and reserves before 2008) and for the Treasury's own marketable debt. Bottom: inflation needed to cover the gap $(\phi^* - \hat{\phi})^+$ after a +1pp permanent rate rise, pp per year for 10 years; bars use the baseline $\hat{\phi}$ (0.39 to 2003, 0 from 2004), the line a common $\hat{\phi} = 0.25$ in every year.

Table 3. Two-layer limit by period (consolidated; g forward; $\psi = 3\text{bp}$)

Period	φ^*	$\hat{\varphi}$	Gap	Inflation-layer ratio	Inflation to cover gap (pp/yr)
1981–85	0.41–0.56	0.39	0.02–0.17	1.4–1.6	0.04–0.23
1986–99	0.12–0.43	0.39	≈ 0	1.3–1.6	≈ 0
2005–07	0.28–0.44	0	0.28–0.44	1.4–1.5	0.38–0.61
2009–21	mostly < 0 (2017–19: 0.11–0.21)	0	≈ 0 (2017–19: 0.11–0.21)	2.0–2.6	≈ 0 (2017–19: 0.23–0.43)
2023–25	0.46–0.48	0	0.46–0.48	2.2–2.4	1.01–1.12

Appendix E reports every year. Three readings follow. First, the fiscal threshold is ordinary today. The period that stands out is 2008–21, when $r + \psi b < g$ and debt stabilized with no fiscal response, and that period is over.

Second, the fiscal response is not ordinary. On the baseline $\hat{\varphi}$, a legislated offset of about 0.39 roughly met the threshold before 2004, and the high-rate 1980s and early 1990s ended in consolidation. Since 2004 the offset has been about zero, so the gap equals the threshold itself. The gap first opened in 2005–07, when φ^* was 0.28–0.44, and the low-rate period of 2008–21 then hid it.

Third, QE raised the price of the inflation substitute. The consolidated ratio rose from 1.2–1.6 in 1980–2007 to 2.0–2.6 from 2008 on. Before QE, consolidation roughly halved the ratio relative to the Treasury's own liabilities (0.49–0.53 times the Treasury-only value in 2003–07), because the Fed held Treasuries against currency, which inflation erodes. Since 2009 the two have been about equal (0.86–1.19), because the Fed's additional Treasuries and MBS are financed by interest-bearing reserves that reprice overnight. The Fed's share of marketable Treasuries was in fact lower at end-2025 (14%) than at end-2007 (16%). What QE changed is how the Fed's book is funded: SOMA Treasuries were 0.88 times currency in 2007 and 1.73 times in 2025, and reserves also fund the MBS.

Decomposition, end-2007 to end-2025. The consolidated ratio rose from 1.45 to 2.18. A chain of counterfactual balance sheets splits the 0.73 rise into three parts. Currency fell from 22% to 8% of interest-bearing liabilities, which adds 0.58. The Treasury's maturity and TIPS choices between 2007 and 2025, with a Fed funded only by currency, subtract 0.23: the Treasury's lengthening worked against the rise. QE, in the sense that reserves rather than currency fund the rest of the Fed's book, adds 0.38.

The QE share depends on the order. Measured first, on the end-2007 balance sheet (the Fed's Treasuries scaled to the end-2025 ratio to currency, and reserves added for the MBS), QE adds 0.57. QE therefore accounts for 52–78% of the rise; the fall of currency relative to debt, which reflects deficits more than monetary policy, accounts for most of the rest.

The combined requirement in 2023–25 is the largest since 1980, about 1.8 times its 2006–07 level.

The equivalent one-time surprise rise in the price level is about 3.3% (3.27–3.42% in 2023–25). As Section 3.4 predicts, it is much less sensitive to consolidation: since 2009 the consolidated value has been 0.95–1.05 times the Treasury-only value.

Robustness. Table 4 recomputes the 2023–25 requirement under alternative parameters, holding the measured clock fixed.

Table 4. Robustness of the 2023–25 requirement (+1pp permanent rate rise, $H = 10$)

Variant	φ^* , 2023–25	Inflation to cover gap, 2023–25 (pp/yr)	Highest before 2022 (year)	2023–25 highest since 1980?
Baseline (forward/SEP g ; $\psi = 3$ bp on consolidated debt)	0.46– 0.48	1.01–1.12	0.61 (2007)	yes
g ex ante: trailing real growth + 10-year expected inflation	0.32– 0.34	0.71–0.80	0.41 (2018)	yes
g : trailing 10-year nominal growth	0.24– 0.28	0.52–0.67	0.63 (2018)	yes
ψ on privately held marketable debt (no reserves)	0.43– 0.45	0.95–1.03	0.61 (2007)	yes
ψ on debt held by the public	0.47– 0.49	1.03–1.14	0.65 (2007)	yes
$\psi = 2$ bp	0.38– 0.40	0.83–0.93	0.57 (2007)	yes
$\psi = 4.5$ bp	0.55– 0.56	1.21–1.33	0.73 (2018)	yes
Monte Carlo: $\psi \sim U[2, 4.5]$ bp, $g \pm N(0, 0.5$ pp), $\hat{\varphi} \sim U[0.25, 0.5]$ to 2003 and $U[0, 0.25]$ after	—	0.36–1.27 (90% band)	—	in 99% of draws

The ranking holds in every variant. Expected growth matters most for the level: an ex-ante measure built from trailing real growth and the Cleveland Fed's ten-year expected inflation puts expected nominal growth at 4.8% in 2025 rather than 3.8%, and lowers the requirement to 0.71–0.80 pp. Where ψ applies barely matters. The debt-rate elasticities in the literature are estimated on debt held by the public, and reserves are zero-duration safe assets; applying ψ only to privately

held marketable debt moves ϕ^* by about 3 points. In the Monte Carlo the 90% band for 2023–25 is 0.36–1.27 pp per year, and 2023–25 is the highest since 1980 in 99% of draws.

Two further checks:

- *Horizon*. With $H = 5$ the 2023–25 requirement is 0.60–0.69 pp per year; with $H = 15$ it is 1.33–1.46.
- *Erosion base*. If all debt counted as erodible and currency were ignored, the requirement would be 1.22–1.41. Ignoring currency alone raises the 2025 consolidated ratio by a third (2.92 against 2.18).

5.3 How large is the fiscal response? Our own estimates

The literature values of $\hat{\phi}$ carry the headline, so we estimate the response ourselves. We regress the federal primary surplus (NIPA, excluding Fed remittances, % of GDP) on one of three lagged variables:

- debt held by the public, as in Bohn (1998);
- interest payments;
- the paper's consolidated debt.

The controls are the output gap, temporary defense spending (defense/GDP minus its HP trend, following Barro and Bohn) and a 2020–21 dummy. The static version is Bohn's. The partial-adjustment version adds the lagged surplus and reports the long-run response. Debt coefficients are mapped to ϕ with the window's mean $r + \psi b$ (Section 5.1).

Recessions both lower the surplus and raise debt, and an output-gap control removes this only in part. We therefore repeat every regression on CBO's primary surplus with the automatic stabilizers removed (fiscal years 1967–2025, as a share of potential GDP; CBO 2026), less Fed remittances, with CBO's GDP gap as the cyclical control.

Table 5. Implied fiscal response ϕ (standard error)

Regressor	Sample	Actual surplus, static	Actual, partial adj.	Cyclically adjusted, static	Cyclically adj., partial adj.
Debt held by the public	1967/71–2003	1.12 (0.51)	1.22 (0.53)	0.99 (0.40)	1.39 (0.69)
	1984–2003	−0.44 (0.44)	0.63 (0.94)	−0.55 (0.39)	2.61 (2.87)
	2004–2024/25	−1.16 (0.09)	−1.23 (0.10)	−1.26 (0.08)	−1.26 (0.09)
	2004–2019	−1.53 (0.17)	−1.71 (0.12)	−1.61 (0.13)	−1.63 (0.07)
Interest payments	1967/71–2003	1.52 (0.51)	2.33 (0.58)	0.77 (0.26)	1.62 (0.69)
	1984–2003	−0.02 (0.45)	1.03 (0.38)	−0.17 (0.38)	1.72 (1.02)
	2004–2024/25	−0.62 (1.41)	−1.95 (1.54)	0.21 (1.67)	−1.63 (1.69)
	2004–2019	3.33 (1.25)	2.94 (1.64)	2.98 (1.12)	2.49 (1.86)
Consolidated debt (this paper)	1984–2003	−0.48 (0.50)	1.02 (1.26)	—	—
	2004–2024/25	−1.05 (0.08)	−1.10 (0.09)	—	—
	2004–2019	−1.42 (0.12)	−1.53 (0.11)	—	—

Newey–West standard errors (2 lags); delta method for the long-run responses. Actual surplus: NIPA, calendar years, from 1971 (debt held by the public from 1970). Cyclically adjusted: CBO, fiscal years, from 1967.

Three things follow.

1. **The estimates agree on direction.** Every debt-based specification is lower after 2004 than before. A break at 2004 is significant for debt held by the public in the static version ($p = 0.005$; $p = 0.002$ cyclically adjusted) and for interest payments with partial adjustment ($p = 0.001$ in both). It is not significant for the consolidated debt measure ($p \approx 0.6$), whose pre-2004 window starts only in 1981 and is itself negative in the static version. The one exception is the interest specification over 2004–19, a period when interest costs were low and falling while the 2011 spending caps cut deficits; its standard error exceeds 1.
2. **They do not agree on size, even without the automatic stabilizers.** Implied ϕ ranges from about -2 to $+3$ across samples and specifications, far outside any plausible structural value. Removing the stabilizers changes the estimates little, so the business cycle is not the main problem. The surplus also moves with revenue booms, tax legislation and wars that are not

responses to debt. That is why Auerbach and Yagan use legislated changes, and we do not replace their estimates with ours.

3. The results that matter do not need the level.

- The inflation layer, and with it the effect of QE, does not depend on $\hat{\phi}$ at all.
- The ranking of 2023–25 as the largest requirement since 1980 survives without assuming a post-2004 collapse. With one common $\hat{\phi}$ in every year, it holds for any $\hat{\phi}$ below about 0.35 (Figure 3, line).
- At $\hat{\phi} = 0.25$ throughout, the 2023–25 requirement is 0.46–0.53 pp per year, against a pre-2022 maximum of 0.43 in 1981.
- Only if the pre-2004 legislated value of 0.39 held in every year does 1981 rank first.

We also tried to identify ϕ with the clock itself, using the predetermined repricing exposure as an instrument for interest-cost changes. The instrument is weak in U.S. annual data (Appendix C). Maturity structures and QE intensities differ widely across countries, so a cross-country panel is the natural way to identify ϕ with this design.

5.4 The level: how much inflation would hold debt stable today?

The combined metric is a sensitivity: inflation per point of a permanent rate shock. A policymaker's question is a level: at today's deficits and today's rates, how much inflation would stop debt/GDP from rising? The same two clocks answer it.

Hold interest-bearing consolidated debt b and the zero-interest base z (currency, and reserves before 2008) constant relative to GDP for H years. The debt-stabilizing primary balance at horizon h is $(\bar{r}(h) - g)b - g \cdot z$:

- the average rate moves from today's $\bar{r}_0$ toward the marginal cost r along the rollover clock, $\bar{r}(h) = \bar{r}_0 + (r - \bar{r}_0)P(h)$;
- the growth of zero-interest money, $g \cdot z$, is seigniorage that finances part of the deficit.

The shortfall against the actual primary balance $s(h)$ must be covered by erosion $\Delta\pi \cdot b \cdot E(h)$:

$$\Delta\pi^{level}(H) = \frac{\int_0^H [(\bar{r}(h) - g)b - g \cdot z - s(h)] dh}{b \int_0^H E(h) dh}.$$

For end-2025, $s(h)$ is CBO's August 2026 baseline primary balance for fiscal years 2026–2035. For 2007 and 2019 we hold the actual primary balance (excluding Fed remittances) constant. The average rate $\bar{r}_0$ prices each privately held security at its own rate, calibrated to the Treasury's official average rate on marketable debt; TIPS add 2% expected inflation, reserves earn the interest rate on reserves, and reverse repos the ON RRP rate.

Table 6. Inflation needed to hold consolidated debt/GDP constant

	End-2007	End-2019	End-2025 (CBO baseline)
Interest-bearing consolidated debt, % of GDP	26	75	95
Zero-interest base (currency; reserves before 2008), % of GDP	5.8	8.4	7.9
Average rate on the stock $\bar{r}_0$, %	4.9	2.3	3.6
Marginal cost r / growth g , %	4.6 / 3.1	2.1 / 3.9	4.2 / 3.8
Primary balance, % of GDP (10-year mean)	+0.97	-2.38	-1.41
Debt-stabilizing primary balance, % of GDP (year 0 $\rightarrow$ year 10)	+0.29 $\rightarrow$ +0.22	-1.52 $\rightarrow$ -1.64	-0.53 $\rightarrow$ +0.03
Shortfall, % of GDP (10-year mean)	-0.74	0.76	1.32
Erosion per pp of inflation, \$bn a year (10-year mean)	18	58	98
Inflation to hold debt/GDP, pp a year, $H = 10$ ($H = 5$)	none	2.8 (2.1)	4.2 (3.0)
Same, discounted at $r - g$	none	2.9	4.1

Five points stand out.

- At end-2025, holding debt/GDP constant through inflation alone would take about **4 pp of extra inflation a year for a decade** (3.0 over five years).
- The primary shortfall averages 1.3% of GDP, about \$400bn a year. Each point of sustained surprise inflation erodes about \$100bn a year on average: more at first, less as the debt reprices.
- The requirement rises with the horizon for the reason in the corollary of Section 3.3: the erodible share of the balance sheet falls from 46% on average over five years to 33% over ten.
- A permanent inflation tax on currency alone would need about 17 pp a year (1.3% of GDP against currency of 7.9% of GDP).
- The same calculation gives 2.8 pp for 2019 and nothing for 2007, when the primary surplus exceeded the debt-stabilizing level.

This number does not use $\hat{\phi}$, since the actual and projected primary balances already embed the fiscal response. It is an order of magnitude, not a forecast:

- It holds the primary balance fixed in real terms. Bracket creep and lags in the indexation of spending would reduce it.
- It holds currency demand and real rates fixed. A flight from currency or a rise in the inflation risk premium would raise it.

5.5 Time to the limit

Table 7 and Figure 4 compute $T(\Delta r)$ at the end-2025 fiscal position: $b = 0.95$, $\bar{r}_0 = 3.6\%$, $r = 4.2\%$, $g = 3.8\%$, $z = 0.079$. The ceilings of 1–3% of GDP bracket the U.S. record. Since 1967 the primary surplus, excluding Fed remittances, has exceeded 2% of GDP only in fiscal years 1997–2001, when it peaked at 4.8%. Each column uses the same fiscal position and a different repricing clock, so the comparison isolates maturity.

Table 7. Years until the debt-stabilizing primary surplus exceeds the ceiling (end-2025 fiscal position)

Surplus ceiling	Rate shock	Consolidated, end-2025	Without QE	End-2007 clock	Consolidated + 2 pp inflation
1% of GDP	+3 pp	0.8	1.5	1.4	2.9
	+4 pp	0.3	0.8	0.8	1.9
	+5 pp	0.2	0.4	0.5	1.2
2% of GDP	+3 pp	4.1	4.8	4.7	6.8
	+4 pp	2.0	2.6	2.4	3.9
	+5 pp	1.1	1.8	1.6	2.5
3% of GDP	+3 pp	never	never	never	never
	+4 pp	5.3	6.2	6.4	8.2
	+5 pp	2.9	3.7	3.6	4.6

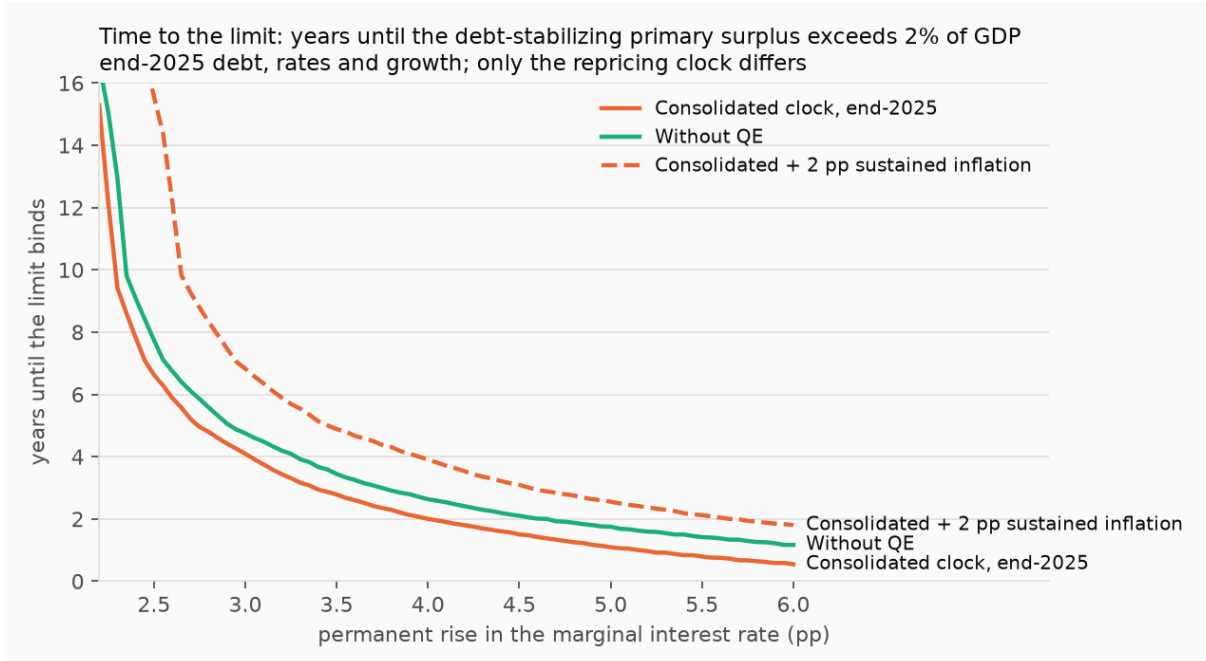


Figure 4. Time to the limit with a ceiling of 2% of GDP on the primary surplus. The end-2025 debt, rates and growth are common to all lines; only the repricing clock differs. The dashed line adds a sustained surprise inflation of 2 pp a year.

Four points stand out.

- With a 2% ceiling, any permanent shock above about 2 pp eventually breaks the limit, whatever the maturity.
- A 3 pp shock breaks it after about four years with today's consolidated clock and after almost five without QE. The faster post-QE clock costs 8–9 months at this ceiling, and more in relative terms for larger shocks: 2.0 against 2.6 years for a 4 pp shock.
- The end-2007 clock gives almost the same times as the no-QE clock. The consolidated structure of 2025 is faster than both.
- Two points of sustained surprise inflation lengthen the time to the limit by about 2–3 years for 3–4 pp shocks, and do not prevent the limit from binding. That is how much time inflation buys.

6. Japan and the United Kingdom

The United States is one case. The United Kingdom and Japan also borrow in their own currencies. Both ran QE on a scale comparable to or larger than the Fed's relative to GDP, and each differs from the United States in a way that matters for the two layers:

- the UK issues the longest debt among the major advanced economies, and about a quarter of it is indexed to inflation;
- Japan has the highest debt, has had r below g in every year of our sample, and until March 2024 paid a zero rate on much of its central-bank reserves.

We build the same consolidated clock, inflation layer and fiscal threshold for both, with the same definitions as for the United States. Interest-bearing liabilities are government securities minus central-bank holdings, plus remunerated reserves as overnight debt. Currency and zero-rate reserves enter only the erosion base, and index-linked debt is not erodible. To make the three countries comparable, all inflation-layer ratios in this section use $g = 4\%$ growth issuance and $H = 10$.

6.1 Data

United Kingdom, 2007–2025.

- Gilts in issue by ISIN at each year-end are from the Debt Management Office's report D1A. Index-linked gilts enter at their inflation-uplifted amount. Undated gilts are treated as perpetual until a redemption date was announced; they were redeemed in 2014–15.
- The security-level sums reproduce the DMO's stated total in all 19 year-ends.
- The Asset Purchase Facility's holdings by gilt are rebuilt from the Bank of England's operation-level results:
 - purchases in 2009–21 and the 2022 financial-stability purchases;
 - active sales from November 2022 and the financial-stability sales.
- Rebuilt this way, the current stock matches the Bank's published holdings table exactly (£405.35bn nominal).
- Reserves are remunerated at Bank Rate and reprice overnight. Notes and coin form the zero-interest base.
- Treasury bills (3–4% of marketable debt) are not yet included.

Japan, fiscal years 2020–2024 (end-March).

- JGBs by issue (general and FILP bonds) are from Table 34 of the Ministry of Finance's debt yearbook. The by-issue sums match the yearbook's totals in each year.
- Bank of Japan holdings are from the BoJ's by-issue release. Government holdings and treasury discount bills are from the BoJ's public-finance statistics; financing bills reprice at three months.
- Floating-rate JGBs reset every six months.
- BoJ current accounts are split by the rate they earn:

- Under the three-tier system of February 2016 to March 2024, the 0% tier joins banknotes in the zero-interest base, and the positive and negative tiers are interest-bearing.
- From the March 2024 reform, only required reserves earn zero.
- Earlier yearbooks are not yet available to us in machine-readable form.

Rates and growth. For the fiscal threshold, r is the marginal cost of each country's existing structure. Each security is priced at the year's average yield for its original tenor: constant-maturity JGB yields for Japan, and Bank of England par yields and SONIA for the UK.

We put g on a common basis across the three countries: trailing 10-year average real growth plus the 2% inflation target that all three central banks share. For the United States this gives $g = 4.4\%$ in 2025, above the FOMC-based 3.8% of Section 5, so the U.S. threshold here is lower than in Table 3. Comparisons in this section are within this common basis.

6.2 The clock and the inflation layer

Table 8. Three sovereigns, own and consolidated ($g = 4\%$, $H = 10$; Japan: fiscal years ending in March of the following year)

	U.S. 2007	U.S. 2025	UK 2007	UK 2021	UK 2025	Japan FY2022	Japan FY2024
Consolidated interest-bearing debt, % of GDP	27	95	32	100	99	150	183
Overnight share of consolidated debt, %	1	14	5	42	22	28	44
Zero-interest base / interest-bearing debt, %	22	8	10	4	3.5	48	11
Index-linked share of consolidated debt, %	11	6	27	22	23	1	0.5
Average maturity, years: own / consolidated	4.6 / 4.7	5.8 / 4.8	14.3 / 13.6	15.0 / 9.4	13.9 / 11.1	8.3 / 7.1	8.6 / 6.0
P(1): own / consolidated	0.38 / 0.37	0.37 / 0.46	0.09 / 0.13	0.10 / 0.48	0.10 / 0.28	0.26 / 0.51	0.24 / 0.59
Inflation-layer ratio $\bar{P}/\bar{E}$: own / consolidated	2.73 / 1.45	2.22 / 2.18	0.90 / 0.81	0.87 / 2.47	0.89 / 1.41	1.20 / 0.81	1.14 / 1.81

Notes. "Own" is the government's own marketable debt, with no central bank. Consolidated debt nets central-bank and government holdings and adds interest-bearing reserves (and, for the United States, reverse repos). The zero-interest base is currency (banknotes, notes and coin) plus reserves that earn nothing: U.S. reserves before October 2008, and Japan's 0% tier and required reserves.

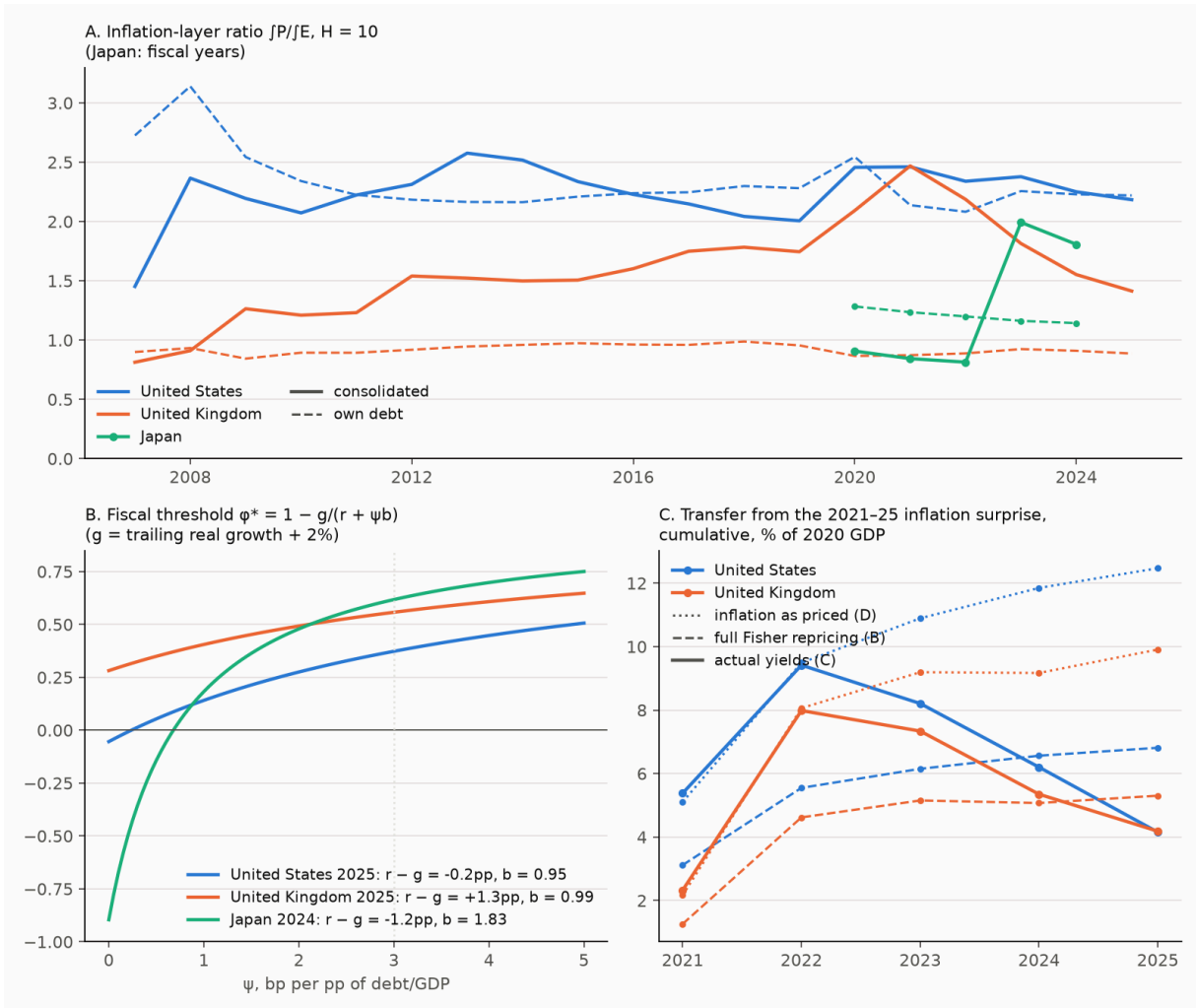


Figure 5. United States (blue), United Kingdom (orange), Japan (green). A: inflation-layer ratio, consolidated (solid) and own debt (dashed), $g = 4\%$, $H = 10$. B: fiscal threshold ϕ^* against the debt-sensitivity of rates ψ , latest year, $g = \text{trailing 10-year real growth} + 2\%$; the dotted line marks the U.S. baseline $\psi = 3\text{bp}$. C: cumulative real transfer from the 2021–25 inflation surprise, from the end-2020 consolidated balance sheet, under full Fisher repricing (B, dashed), inflation as priced (D, dotted) and actual yields (C, solid).

Figure 5A shows the ratios over time. Four findings follow.

(i) On their own debt, the three governments look very different. The UK's gilts have an average maturity of 14–16 years, and only about a tenth of a rate shock reaches the average rate within a year. Japan is in between, and U.S. Treasury debt reprices fastest. Because long debt is exposed to inflation for longer, and index-linked gilts are exposed to none, the inflation needed per unit of relief on own debt is lowest in the UK (about 0.9), then Japan (1.1–1.3), then the United States (2.1–2.7). Judged by debt management alone, the UK is the best placed of the three to inflate.

(ii) Consolidation erases most of the difference. At the QE peak the consolidated ratios were almost equal: 2.47 for the UK at end-2021, 2.46 for the United States at end-2021, and 1.99 for Japan at end-March 2024. The APF held 36% of gilts at end-2021 and financed them with reserves paid Bank Rate. That cut the average maturity of the UK's consolidated liabilities from 15.0 to 9.4 years and put 42% of them at the overnight rate. The UK's long issuance was, for the purpose of the inflation layer, largely undone by the central bank.

(iii) What matters is how the central bank funds its bonds. Before QE, consolidation lowered the ratio in both the United States (2.73 to 1.45) and the UK (0.90 to 0.81), because the central bank's liabilities were mostly currency, which inflation erodes. Japan shows the same mechanism at scale. In FY2022 the BoJ held 47% of JGBs, but a large part of its current accounts sat in the 0% tier. Together with banknotes, the zero-interest base was 48% of interest-bearing consolidated debt, and the consolidated ratio, 0.81, was below Japan's own-debt ratio.

The March 2024 reform ended negative rates and moved all but required reserves to the policy rate. With almost no change in the government's own debt, the consolidated ratio rose from 0.81 at end-March 2023 to 1.99 at end-March 2024, and interest-bearing consolidated debt rose from 150% to 194% of GDP. This is the counterfactual of Section 7 run in reverse: paying interest on reserves takes them out of the inflation-tax base. Japan's reform did so at a stroke. The rise, 1.18, exceeds the entire rise in the U.S. consolidated ratio from 2007 to 2021 (1.45 to 2.46).

(iv) QT unwinds the effect, and faster in the UK. Active gilt sales and redemptions took the APF from 36% of gilts in 2021 to 17% in 2025. Over the same period the UK consolidated ratio fell from 2.47 to 1.41, and the overnight share from 42% to 22%, while the own-debt clock did not move. The Fed, which ran off its holdings without sales, ended 2025 with a ratio of 2.18.

6.3 The fiscal threshold

Table 9. Fiscal threshold and inflation requirement on a common basis (+1pp permanent rate rise, $H = 10$)

	U.S. 2025	UK 2025	Japan FY2024
Marginal cost r / growth g , %	4.2 / 4.4	4.8 / 3.4	1.4 / 2.6
$r - g$, pp	-0.2	+1.3	-1.2
Consolidated debt b , % of GDP	95	99	183
φ^* at $\psi = 0 / 1 / 3 / 4.5$ bp	-0.05 / 0.14 / 0.37 / 0.48	0.28 / 0.41 / 0.56 / 0.63	-0.90 / 0.18 / 0.62 / 0.73
φ^* at $\psi = 3$ bp, trailing nominal g	0.24	0.39	0.72
Required inflation, pp a year, $\psi = 3$ bp: $\hat{\varphi} = 0$ / 0.25	0.82 / 0.27	0.79 / 0.44	1.12 / 0.66
Required inflation, pp a year, $\psi = 0$: $\hat{\varphi} = 0$ / 0.25	0 / 0	0.40 / 0.05	0 / 0

Figure 5B plots φ^* against ψ . Three findings follow.

1. **The UK is the only one of the three with r above g .** In 2025 its marginal cost exceeded trend nominal growth by 1.3 pp. Its fiscal threshold is therefore positive even if rates do not respond to debt ($\psi = 0$): the surplus must offset at least 28% of the marginal interest cost. At $\psi = 0$ and with no fiscal response, the UK is the only country where the inflation layer is needed at all.
2. **Japan's threshold depends almost entirely on ψ .** With r 1.2 pp below g and debt at 183% of GDP, Japan's threshold is -0.90 at $\psi = 0$ and 0.62 at $\psi = 3$ bp, higher than either of the others. The debt-sensitivity of rates estimated for the United States cannot be assumed for Japan, where yields stayed near zero while debt rose past twice GDP. Japan's distance to the limit is therefore mainly a question about ψ , which its own history of yield control does not identify.
3. **The United States sits on the boundary.** At $r \approx g$, the U.S. threshold is close to zero without a debt premium, and 0.37 at $\psi = 3$ bp on this basis (0.47 with the FOMC growth path of Section 5).

At $\psi = 3$ bp and no fiscal response, covering the fiscal gap after a permanent 1pp rate rise would take about 0.8 pp of extra inflation a year for a decade in both the United States and the UK, and 1.1 pp in Japan (Table 9). The countries reach similar numbers by different routes:

- the UK has the higher threshold but the lower ratio, because of QT and its index-linked gilts;

- Japan has the highest threshold and, since 2024, a ratio close to the others.

6.4 The 2021–25 inflation surprise in the United Kingdom

We repeat the test of Section 4.5 on the UK's end-2020 consolidated balance sheet:

- privately held conventional gilts, £871bn;
- reserves, £770bn;
- notes and coin, £92bn.

Index-linked gilts (£454bn, privately held) carry no nominal claim to erode and are reported separately. Expected CPI inflation is the end-2020 five-year implied RPI inflation (3.16%) less the average RPI–CPI wedge of 2011–20 (0.77 pp), that is 2.39%. The surprise is realized CPI inflation minus 2.39.

The design differs from the U.S. test in one respect: we hold the end-2020 portfolio closed, so new borrowing and QT after 2020 are left out. Treasury bills are also excluded. Each gilt is rolled at redemption into a gilt of the same original tenor, and amounts stay at their end-2020 levels. Yields are Bank of England zero-coupon nominal, real and implied-inflation curves at 5, 10 and 20 years, with Bank Rate at the short end.

Table 10. United Kingdom: real transfer to the consolidated government from the 2021–25 inflation surprise (cumulative, % of 2020 GDP)

End of	Cumulative surprise (pp)	Gross erosion	Model: full Fisher repricing (B)	Model, analytic s·b·E	Inflation as priced (D)	Actual (C)	Memo: extra interest on reserves in C	Memo: RPI surprise on index-linked gilts
2021	2.9	2.3	1.3	1.2	2.2	2.3	0.0	0.9
2022	10.5	8.5	4.6	4.2	8.1	8.0	0.5	2.9
2023	11.9	9.7	5.2	4.7	9.2	7.3	2.2	3.3
2024	12.1	9.9	5.1	4.8	9.2	5.4	4.1	3.4
2025	13.0	10.6	5.3	5.0	9.9	4.2	5.7	3.6

Notes. Extra interest on reserves is measured relative to the no-surprise forward path. The RPI surprise on index-linked gilts is their inflation uplift in excess of the end-2020 implied RPI inflation. It is a nominal cost, not a real transfer, because holders of those gilts are fully compensated.

The UK results repeat the U.S. pattern closely (Figure 5C).

1. **Markets repriced far less than full Fisher repricing implies.** With inflation compensation as actually priced (D), the transfer reached 9.2% of GDP by 2023, 1.8 times the 5.2% under full Fisher repricing (B). This is the same multiple as in the United States. The analytic layer $s \cdot b \cdot E$ again tracks its numerical counterpart B.
2. **Higher real rates took most of it back, and in the UK almost all of that ran through reserves.** With actual yields (C), the transfer peaked at 8.0% of GDP in 2022 and fell to 4.2% by 2025, the same end-point as in the United States. The gap between C and D in 2025, 5.7% of GDP, is about equal to the extra interest paid on reserves as Bank Rate rose to 5.25%. Reserves made up 47% of the UK's interest-bearing nominal liabilities at end-2020. The rate layer that reversed the inflation transfer was, in the UK, the cost of the APF's overnight funding.
3. **Index-linked gilts shifted part of the inflation to the government.** RPI inflation exceeded its end-2020 implied rate by 16.6 pp over 2021–25. The resulting uplift on privately held index-linked gilts added 3.6% of GDP to nominal debt, about a third of the gross erosion of nominal liabilities. This is the other face of the UK's low own-debt ratio: debt that inflation cannot erode also cannot share in an inflation surprise.

6.5 *What the comparison adds*

Three points generalize beyond the United States.

- **The inflation layer is a property of the consolidated balance sheet, not of debt management.** The UK issued the longest debt of the three and ended 2021 with the same consolidated ratio as the United States.
- **The funding of the central bank's bonds is the decisive margin.** Japan's zero-rate reserves kept its ratio below one, and the end of that regime raised it by more than the U.S. ratio rose from 2007 to 2021.
- **The fiscal threshold is where the countries differ.** Here what matters is $r - g$ and the debt-sensitivity of rates, which for Japan is the open question.

Three sovereigns are not a panel. They do show, however, that the variation in maturity, indexation and central-bank funding needed to identify the fiscal response across countries exists.

7. Counterfactuals and policy options

Table 11 changes one element of the end-2025 balance sheet at a time and reports the combined metric. Unless stated, debt/GDP is held at its baseline so that each scenario isolates a change in composition.

Table 11. End-2025 counterfactuals (+1pp permanent rate rise, $H = 10$; baseline $\phi = 0$)

Scenario	ϕ^*	Gap	Inflation to cover gap (pp/yr)	Change vs. baseline	One-time jump (%)
Baseline (consolidated; $g = 3.8\%$, SEP)	0.46	0.46	1.00	—	3.26
<i>Fiscal policy</i>					
Fiscal response restored ($\phi = 0.39$)	0.46	0.07	0.16	−84%	0.51
Fiscal response 0.25 (intermediate)	0.46	0.21	0.46	−54%	1.50
<i>Central-bank balance sheet</i>					
No QE: Fed Treasuries = currency, no reserves	0.46	0.46	0.83	−18%	3.07
QT: reserves to \$1.9tn, Treasuries back to private	0.46	0.46	0.94	−6%	3.20
Ending interest on reserves	0.46	0.46	0.67	−33%	2.82
Tiering: 50% of reserves unremunerated	0.46	0.46	0.82	−19%	3.04
Tiering: 25% of reserves unremunerated	0.46	0.46	0.90	−10%	3.15
Fed Treasury portfolio 50% bills (from 5.5%)	0.46	0.46	0.89	−12%	3.14
<i>Treasury debt management</i>					
Bill share 25% of marketable debt (from 21.6%)	0.46	0.46	1.06	+6%	3.31
Bill share 30%	0.46	0.46	1.15	+15%	3.39
Terms out: 10% of bills → 10-year notes	0.46	0.46	0.93	−7%	3.18
<i>Markets and macro</i>					
Convenience yield −50bp	0.50	0.50	1.08	+8%	3.51
Convenience yield −100bp	0.53	0.53	1.15	+14%	3.73
Trend growth 4%	0.43	0.43	0.95	−5%	3.07
Trend growth 3.5%	0.50	0.50	1.08	+8%	3.55
$\psi = 2\text{bp}$ (CBO)	0.38	0.38	0.82	−18%	2.67
$\psi = 4.5\text{bp}$	0.55	0.55	1.20	+20%	3.90
<i>Accounting benchmark</i>					
Treasury's own liabilities (Fed ignored)	0.46	0.46	1.02	+1%	3.31

The scenarios are defined as follows.

- *No QE*. The Fed holds only as many Treasuries as it has currency outstanding, as before 2008. The rest return to private holders pro rata to the SOMA maturity structure, and reserves and reverse repos go to zero.
- *Ending interest on reserves; tiering*. All, half or a quarter of the \$2.85tn of reserves stop earning interest. They move from overnight interest-bearing debt to the zero-interest base that inflation erodes.
- *Fed Treasury portfolio 50% bills*. The Fed sells coupons pro rata to private holders and buys bills from them until half its Treasuries are bills, a change of about \$1.9tn.
- *Bill share*. Treasury raises bills to 25% or 30% of marketable debt and shrinks all other securities pro rata.

Five results follow.

1. **The fiscal response is the dominant lever.** Restoring the pre-2004 response cuts the requirement by 84%. No balance-sheet policy comes close.
2. **Balance-sheet policy moves only the inflation layer.** None of the central-bank or debt-management scenarios changes ϕ^* when debt/GDP is held fixed. They change the cost of the inflation substitute by 6–33%.
3. **Ending interest on reserves is the largest balance-sheet lever.**
 - It cuts the requirement by a third (36% once debt/GDP falls with the interest-bearing stock), almost twice the effect of undoing QE. Tiering at 50% or 25% cuts it by 19% or 10%.
 - In levels (Section 5.4), it lowers the inflation needed to hold debt/GDP stable from 4.2 to 2.3 pp a year, because reserves stop costing interest (about \$104bn a year at the end-2025 rate of 3.65%) and their growth becomes seigniorage.
 - It is not a free lunch. Unremunerated reserves are a tax on banks. In an ample-reserves framework the interest rate on reserves is also how the Fed sets the policy rate, so ending it would require far smaller reserves or binding reserve requirements. The ECB's move to stop remunerating minimum reserves in 2023 is a partial version of the same idea.
4. **A Fed shift to bills lengthens the consolidated structure.** Moving half the Fed's Treasuries into bills cuts the requirement by 12%. The Fed's own portfolio gets shorter, but the private sector ends up holding more coupons, and it is the private sector's holdings that matter. From the consolidated point of view, the Fed buying bills is Treasury terming out.
5. **More bills raise the price of the inflation substitute.** A bill share of 25% raises the requirement by 6%, and 30% by 15%. Terming out a tenth of bills lowers it by 7%.

When debt/GDP is allowed to change with the interest-bearing stock, the results are similar:

- No QE without the reserves that fund the Fed's MBS lowers b from 0.95 to 0.90 and the requirement by 20%.
- Ending interest on reserves lowers b to 0.86, ϕ^* to 0.44 and the requirement by 36%.
- Tiering at 50% lowers the requirement by 20%.

Losing the convenience yield, slower growth and a steeper debt-rate schedule are different in kind: they raise the threshold itself.

Two limitations apply. The balance-sheet scenarios hold r fixed and ignore the term-premium effects of QE, QT and changes in bill supply. And all scenarios are static comparisons.

8. Conclusion

A reserve-currency sovereign does not run into a debt wall. It runs into a choice between fiscal adjustment and an inflation tax.

The size of the required adjustment, the fiscal layer, depends on $r - g$, on how sensitive rates are to debt and on the debt level, not on maturity. How expensive the inflation alternative is, the inflation layer, depends on how fast the consolidated liabilities reprice and on how much of them inflation can erode. Both are measurable. QE made the clock faster and, by replacing the Fed's currency funding with interest-bearing reserves, shrank the base the inflation tax falls on.

The United States in 2023–25 faces an ordinary fiscal threshold with an unusually absent fiscal response and an unusually fast consolidated clock. Inflation would therefore buy less relief per point than at any time since 1980. Inflation on debt only buys time, and a permanent inflation tax on currency alone would take about 5.6 pp a year per point of rate shock. In levels, holding debt/GDP stable at CBO's baseline deficits would take about 4 pp of extra inflation a year for a decade. When the primary surplus has a ceiling, maturity decides how long the limit takes to bind: a 3 pp permanent shock breaks a ceiling of 2% of GDP in about four years at today's consolidated clock, and two points of sustained inflation buy two to three more. The durable resolutions are a return of the fiscal response that met the threshold before 2004, or a one-time surprise revaluation of the debt.

Several issues remain open. The level of the fiscal response is not well identified: its post-2004 collapse rests mainly on one study, and our own regressions agree on its direction but not its size. The main ranking holds for any constant response below 0.35, and a cross-country panel, with its wider variation in maturity and QE, is the natural way to identify it. Expected growth is the parameter that matters most for the level of the requirement; an ex-ante measure lowers it by about 30%, although the ranking survives. The balance-sheet counterfactuals abstract from term-premium effects, and the combined metric is a first-order, undiscounted composition. Finally,

currency demand is held fixed; a sustained inflation tax would shrink it, so the requirements are lower bounds in that respect.

The comparison with the United Kingdom and Japan suggests that these conclusions are not specific to the United States. In all three countries QE funded by interest-bearing reserves made the inflation substitute costlier, and QT or zero-rate reserves made it cheaper again. The UK's long, partly indexed debt did not protect its consolidated balance sheet from that effect. Extending the Japanese series before 2020 and adding the UK's Treasury bills are the next steps.

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Appendix A. Data construction details

Repricing distribution. For each year-end t and each liability i with par (or balance) w_i and years to repricing τ_i , the repricing distribution is $F_t(h) = \sum_i w_i \cdot 1\{\tau_i \leq h\} / \sum_i w_i$. Repricing dates are:

- maturity for bills, notes, bonds and TIPS;
- one week for floating-rate notes (weekly reset);
- zero (overnight) for reserve balances and reverse repurchase agreements.

The clock adds growth-financing issuance at the new rate: $P_t(h) = 1 - (1 - F_t(h))e^{-gh}$, with $g = 4\%$ for the descriptive clock (Figure 1) and the year's g in the limit calculations. The erosion clock P_N is built the same way from non-indexed liabilities only.

Zero-interest base. Currency in circulation from H.4.1 (series RESTBC), in the same week as reserves, from 2003. Before that, December averages of currency in circulation and reserve balances (FRED CURRCIR and RESBALNS). The two currency sources differ by about 1% where they overlap. Reserves are in the zero-interest base through 2007 and in overnight interest-bearing debt from 2008.

Treasury-only stock. All unmatured marketable Treasuries in MSPD Table 3 at the year-end, one row per CUSIP, including those held by the Federal Reserve.

Consolidated stock. Marketable Treasuries minus SOMA holdings, matched by CUSIP (with TIPS inflation compensation), plus reserve balances and reverse repos from the H.4.1 week closest to, and not after, the year-end. SOMA CUSIPs that are absent from the MSPD year-end snapshot all mature between the SOMA date and 31 December (at most \$61bn, 0.3% of the stock) and are dropped.

1980–2002. FD-5 reports privately held marketable debt in five remaining-maturity buckets: within 1 year, 1–5, 5–10, 10–20 and 20 years and over. Within each bucket we spread par uniformly over the bucket (upper edge of the last bucket: 30 years). On the exact security-level

data for 2003–07, this bucketing understates the ten-year inflation-layer ratio $\bar{P}/\bar{E}$ by 0.077 (s.d. 0.016), $\bar{P}$ by 0.167, and the remaining-maturity rate r by 0.28 pp. We add these corrections to the 1980–2002 values. TIPS (1997–2002) are removed from the erosion base by bucket: their December totals come from Treasury Bulletin table FD-2 and their maturity profile from the TIPS issues outstanding at each year-end.

Required inflation. The integrals in Result 3 are evaluated by the trapezoid rule on a grid with a step of one month over $[0, 15]$ years. $F(h)$ is interpolated between the sorted repricing dates of the security-level data.

Backtest projection (Section 4.3).

- From the year-end portfolio, every surviving security keeps its rate: the coupon for notes and bonds, the real coupon for TIPS, and the 3-month rate plus spread for FRNs.
- Each month, maturing principal plus net new borrowing is reissued in year t 's gross auction mix, at that month's H.15 yield for the tenor.
- The projected average rate is the par-weighted rate of the resulting portfolio. We score its change against the change in FiscalData's official average interest rate on total marketable debt.
- "Realized borrowing" interpolates the actual path of marketable debt. "Trend borrowing" grows the year- t stock at 4% a year.

Fiscal-reaction coefficient. Auerbach and Yagan (2024) estimate the legislated primary-surplus response to lagged net interest. Their coefficient is in dollars of surplus per dollar of interest, which is the marginal offset ϕ of Section 3. Eichengreen, Menuet and Donnat (2026) instead estimate a log-log elasticity of the surplus to debt service. That elasticity maps to ϕ only through $s/(\bar{r}b)$, which is undefined when the primary balance is negative, so we use it as qualitative evidence only.

United Kingdom (Section 6).

- *Gilts.* We use the DMO's report D1A for the last business day of each year, 2007–2025. Conventional gilts, including "rump" gilts, enter at their nominal amount. Index-linked gilts, with both the 3-month and the 8-month indexation lag, enter at their nominal amount including the inflation uplift. Undated gilts have no repricing date and are given one of 100 years until the Treasury announced their redemption in 2014–15. The parsed sums reproduce the total stated in each report.
- *APF holdings.* Holdings of each gilt at any date are the sum of the nominal amounts in all operations settled by that date. A holding drops out at the gilt's redemption, which is read from the Bank's bond code, and index-linked holdings are uplifted with the DMO ratio of amount to nominal for the same gilt. The operations are:

- APF purchases, 2009–21, including reinvestments;
- the 2022 financial-stability purchases of long-dated conventional and index-linked gilts;
- minus active sales from November 2022 and the financial-stability sales of November 2022 to January 2023.
- *Other liabilities and GDP.* Reserve balances and notes and coin are the last December observations of Bank of England series LPMBL22 and LPMAVAA. GDP is the calendar-year sum of quarterly nominal GDP (FRED UKNGDP).
- *Marginal cost.* The marginal cost r prices each gilt at the calendar-year average yield for its original tenor (redemption date minus first issue date). Yields are interpolated between SONIA and the Bank's 5-, 10- and 20-year nominal par yields, flat beyond 20 years; undated gilts are priced at 20 years.
- *Growth.* Real growth for g is from FRED NGDPRSAXDCGBQ.
- *Inflation test.* The test in Section 6.4 uses the Bank's 5-, 10- and 20-year zero-coupon nominal, real and implied-inflation curves (monthly averages), with Bank Rate at zero maturity and the real curve flat below 5 years. Inflation is from the ONS series D7BT (CPI) and CHAW (RPI).

Japan (Section 6).

- *JGBs by issue.* JGBs by issue at each fiscal year-end come from Table 34 of the Ministry of Finance's debt yearbook (general bonds and FILP bonds). Columns are located by their header labels, because the layout differs across editions. Issues are classified as:
 - index-linked (物価連動);
 - floating-rate (変動), which reprice at the next six-month reset;
 - discount bills (割引);
 - fixed-rate.
- *Holdings.* BoJ holdings are matched by issue name and number from the BoJ's release of JGB holdings by issue at the fiscal year-end (the last business day of March).
- *Financing bills and holder shares.* Financing bills are the treasury-discount-bill total in the BoJ's public-finance statistics (PF02) minus the treasury bills in Table 34, and reprice at three months. BoJ holdings of bills (balance-sheet item MABJMA5A) and government holdings of bills and JGBs (PF02) are allocated pro rata.
- *Current accounts.*
 - Current accounts are the end-March balance (MABJML11).
 - Through February 2024 the 0% tier's share of current accounts is the March average share of balances at a zero rate among the three tiers (MD08). We apply that share to

the end-March balance.

- After the March 2024 reform the zero-rate part is required reserves (MD07, average outstanding).
- Banknotes (MABJML1) complete the zero-interest base.
- *GDP*. GDP is the fiscal-year average of quarterly nominal GDP at annual rates (FRED JPNNGDP).
- *Marginal cost*. The marginal cost r prices each issue at the fiscal-year average of the Ministry of Finance's constant-maturity yield for its original tenor. Bills and floating-rate bonds are priced at one year, and index-linked bonds at the 10-year nominal yield.
- *Growth*. Real growth for g is the fiscal-year growth of FRED JPNRGDPEXP.

Appendix B. Proofs

B.1 Repricing law. Let B be the face value of debt and $R = \bar{r}B$ total interest. Debt matures at rate λ ; the average rate on the maturing flow is $\bar{r}$ (exact under exponential maturity). Maturing debt and net issuance $\dot{B}$ are refinanced at the market rate r , so $\dot{R} = (\lambda B + \dot{B})r - \lambda B \cdot \bar{r}$. Then

$$\dot{\bar{r}} = \frac{\dot{R}}{B} - \bar{r} \frac{\dot{B}}{B} = \left(\lambda + \frac{\dot{B}}{B} \right) (r - \bar{r}).$$

A cohort-level simulation reproduces this law to within 4×10^{-7} , which is the time-step error.

B.2 Stability with exponential repricing. Take the state $(b, \bar{r})$ and let $\Lambda = \lambda + g$. Around the steady state (b^*, r^*) ,

$$J = \begin{pmatrix} (1 - \phi)r^* - g & (1 - \phi)b^* \\ \psi\Lambda & -\Lambda \end{pmatrix}, \quad \det J = \Lambda [g - (1 - \phi)(r^* + \psi b^*)].$$

Write $a = (1 - \phi)r^* - g$ and $c = (1 - \phi)b^* \geq 0$ for $\phi \leq 1$. Then $\det J > 0$ if and only if $a + c\psi < 0$. That implies $a < 0$, and therefore $\text{tr } J = a - \Lambda < 0$. So the steady state is locally stable if and only if $(1 - \phi)(r^* + \psi b^*) < g$, whatever Λ is.

B.3 Any repricing kernel. Let the average rate be a weighted history of market rates, $\bar{r}(t) = \int_0^\infty k(s) r(t - s) ds$, with $k \geq 0$ and $\int k = 1$. The cumulative kernel $\int_0^h k$ is the rollover clock $P(h)$. Linearizing gives $\dot{b} = a \cdot b + c \cdot \bar{r}$ with $r = \psi b$, so modes e^{xt} solve

$$x = a + c\psi K(x), \quad K(x) = \int_0^\infty k(s) e^{-xs} ds.$$

- If $\text{Re } x \geq 0$, then $|K(x)| \leq 1$, so $\text{Re } x = a + c\psi \text{Re } K(x) \leq a + c\psi$. Under the stability condition, $a + c\psi < 0$. This is a contradiction, so every root has $\text{Re } x < 0$.
- Conversely, if $a + c\psi > 0$, the real function $f(x) = a + c\psi K(x) - x$ is positive at $x = 0$ and tends to $-\infty$. It therefore has a positive real root.

Stability thus depends on (a, c, ψ) and not on the kernel. The exponential kernel $k(s) = \Lambda e^{-\Lambda s}$ gives $K(x) = \Lambda/(\Lambda + x)$ and reproduces B.2.

With lagged fiscal adjustment, $\dot{s} = \kappa[s^* + \varphi(\bar{r}b - r^*b^*) - s]$, the determinant condition is the same. A scan of 3,180 combinations of (r, φ, κ) with $\lambda \in [0.05, 5]$ finds no case in which λ changes stability.

B.4 Required inflation (Result 3). A permanent real-rate rise Δr raises interest by $\Delta r \cdot b \cdot P(h)$ at horizon h , where P is the clock of all interest-bearing debt $b = N + I$ (nominal plus indexed).

A sustained surprise inflation $\Delta\pi$ lowers the real value of a nominal liability until it reprices, since repriced debt carries a Fisher-adjusted coupon. Indexed debt I is not eroded at all. Currency C pays no interest and never reprices, so it is eroded at every h . The erosion flow at h is therefore $\Delta\pi \cdot \{N[1 - P_N(h)] + C\} = \Delta\pi \cdot b \cdot E(h)$. Setting cumulative erosion over $[0, H]$ equal to the unfinanced share u of cumulative extra interest,

$$\Delta\pi b \int_0^H E = u \Delta r b \int_0^H P \Rightarrow \Delta\pi^{req}(H) = u \Delta r \frac{\int_0^H P}{\int_0^H E}.$$

With $u = (\varphi^* - \varphi)^+$ this is the combined metric of Section 3.3.

Three remarks:

1. φ^* is evaluated at r rather than $r + \Delta r$. Since φ^* rises with r , the metric is conservative, and the error is second order in Δr .
2. Since $\int_0^\infty (1 - P_N) = \int_0^\infty (1 - F_N) e^{-gh} dh < \infty$, both $\int_0^H P$ and $\int_0^H E$ grow like H , with slopes 1 and C/b . The required rate therefore converges to $u \cdot \Delta r \cdot b / C$ as $H \rightarrow \infty$. Without currency it diverges.
3. For a one-time level jump Δp , erosion is $(N + C) \cdot \Delta p$ regardless of maturity. So $\Delta p^{req} = u \cdot \Delta r \cdot \int_0^H P / (N/b + C/b)$.

Appendix C. Identifying φ with the repricing exposure

We tried to estimate φ directly with the clock.

- **Instrument.** The predicted interest-cost change $b_t \cdot P_t(1) \cdot \Delta r_{t+1}$ uses the predetermined repricing exposure.
- **Outcome.** The subsequent change in the NIPA primary surplus, excluding Fed remittances.
- **Controls.** The rate change, debt, growth and the lagged surplus.

In annual data for 1980–2024 the instrument is weak:

- First-stage $F \approx 6.7$, falling below 1 once $\Delta r \cdot b$ is controlled.

- The implied ϕ is implausible (6–26).

The reason is that the U.S. repricing speed varies too little over time. $P(1)$ has a standard deviation of 0.056, and the instrument is 99% correlated with $\Delta r \cdot b$.

The design needs cross-sectional variation in repricing speed, which a panel of countries would supply.

Appendix D. Reproduction

The replication package is at <https://github.com/dingningpei/rollover-clock>. It contains the code (`src/`) and three hand-collected inputs (`data/manual/`):

- the Treasury Bulletin FD-5/FD-7 December rows for 1980–2003, transcribed from FRASER scans;
- the Federal Reserve Banks' combined income statements for 2022–2025;
- the FOMC's December longer-run growth projections.

Everything else is downloaded from public sources: the Treasury (FiscalData: MSPD, auctions, average interest rates), the Federal Reserve Board (H.4.1 and H.15 bulk files), the New York Fed (SOMA holdings, EFFR) and the BEA (NIPA). The command

```
bash reproduce.sh
```

downloads the inputs, runs the model checks and the full pipeline, and writes all tables to `data/processed/` and Figures 1–3 to `paper/figures/`.

Section 6 uses the code in `src/intl/`. The DMO's gilts-in-issue reports (D1A, 2007–2025) are included in `data/manual/uk_dmo/`, because the DMO site does not allow scripted downloads. The other inputs are downloaded from public sources:

- the Bank of England (APF operation results and holdings table, statistical database);
- the ONS (CPI and RPI);
- Japan's Ministry of Finance (debt yearbook, constant-maturity JGB yields);
- the Bank of Japan (JGB holdings by issue, time-series API);
- FRED (GDP).

`src/intl/comparison.py` writes Tables 8–9 and Figure 5; `src/intl/uk_inflation_test.py` writes Table 10.

Appendix E. The two-layer limit by year, 1980–2025

Table E1. Two-layer limit by year (consolidated; +1pp permanent rate rise, $H = 10$; $\psi = 3\text{bp}$; g forward to 2015, SEP from 2016)

Year	Source	r (%)	g (%)	b	ϕ^*	$\hat{\phi}$	Gap	Ratio, consolidated	Ratio, Treasury	Inflation to cover gap (pp/yr)	One- time jump (%)
1980	FD-5	12.04	7.66	0.17	0.39	0.39	0.00	1.34	—	0.00	0.00
1981	FD-5	14.76	6.76	0.18	0.56	0.39	0.17	1.39	—	0.23	0.98
1982	FD-5	12.91	6.93	0.22	0.49	0.39	0.10	1.52	—	0.15	0.60
1983	FD-5	10.49	6.58	0.25	0.41	0.39	0.02	1.54	—	0.04	0.15
1984	FD-5	11.87	6.09	0.27	0.52	0.39	0.13	1.56	—	0.20	0.79
1985	FD-5	9.70	5.83	0.29	0.45	0.39	0.06	1.50	—	0.09	0.35
1986	FD-5	7.30	5.84	0.30	0.29	0.39	0.00	1.42	—	0.00	0.00
1987	FD-5	7.86	5.86	0.31	0.33	0.39	0.00	1.35	—	0.00	0.00
1988	FD-5	8.51	5.64	0.30	0.40	0.39	0.01	1.33	—	0.01	0.06
1989	FD-5	8.81	5.50	0.30	0.43	0.39	0.04	1.32	—	0.06	0.25
1990	FD-5	8.50	5.57	0.32	0.41	0.39	0.02	1.37	—	0.03	0.13
1991	FD-5	7.09	5.57	0.35	0.32	0.39	0.00	1.42	—	0.00	0.00
1992	FD-5	5.63	5.31	0.37	0.21	0.39	0.00	1.46	—	0.00	0.00
1993	FD-5	4.82	5.27	0.38	0.12	0.39	0.00	1.49	—	0.00	0.00
1994	FD-5	6.46	5.31	0.38	0.30	0.39	0.00	1.49	—	0.00	0.00
1995	FD-5	6.54	5.50	0.38	0.28	0.39	0.00	1.56	—	0.00	0.00
1996	FD-5	6.24	5.53	0.38	0.25	0.39	0.00	1.55	—	0.00	0.00
1997	FD-5	6.32	5.38	0.35	0.27	0.39	0.00	1.48	—	0.00	0.00
1998	FD-5	5.47	5.02	0.32	0.22	0.39	0.00	1.41	—	0.00	0.00
1999	FD-5	5.72	4.19	0.29	0.36	0.39	0.00	1.33	—	0.00	0.00
2000	FD-5	6.42	3.94	0.24	0.45	0.39	0.06	1.23	—	0.07	0.31
2001	FD-5	4.48	3.99	0.23	0.23	0.39	0.00	1.23	2.47	0.00	0.00
2002	FD-5	3.48	4.08	0.24	0.03	0.39	0.00	1.27	2.65	0.00	0.00
2003	MSPD+SOMA	2.75	3.98	0.26	-0.13	0.39	0.00	1.32	2.71	0.00	0.00
2004	MSPD+SOMA	3.17	3.75	0.27	0.06	0.00	0.06	1.37	2.71	0.08	0.35
2005	MSPD+SOMA	4.01	3.47	0.27	0.28	0.00	0.28	1.37	2.74	0.38	1.67
2006	MSPD+SOMA	4.84	3.15	0.26	0.44	0.00	0.44	1.37	2.71	0.60	2.65
2007	MSPD+SOMA	4.56	3.10	0.26	0.42	0.00	0.42	1.45	2.73	0.61	2.61
2008	MSPD+SOMA	2.69	3.43	0.42	0.13	0.00	0.13	2.37	3.14	0.32	0.98
2009	MSPD+SOMA	1.99	4.05	0.52	-0.14	0.00	0.00	2.19	2.54	0.00	0.00

Year	Source	r (%)	g (%)	b	ϕ^*	$\hat{\phi}$	Gap	Ratio, consolidated	Ratio, Treasury	Inflation to cover gap (pp/yr)	One- time jump (%)
2010	MSPD+SOMA	1.94	3.58	0.59	0.04	0.00	0.04	2.07	2.34	0.07	0.25
2011	MSPD+SOMA	1.73	4.32	0.63	-0.19	0.00	0.00	2.22	2.23	0.00	0.00
2012	MSPD+SOMA	1.14	4.88	0.68	-0.54	0.00	0.00	2.31	2.18	0.00	0.00
2013	MSPD+SOMA	1.55	5.17	0.73	-0.39	0.00	0.00	2.58	2.16	0.00	0.00
2014	MSPD+SOMA	1.83	5.27	0.73	-0.31	0.00	0.00	2.52	2.16	0.00	0.00
2015	MSPD+SOMA	1.67	5.38	0.73	-0.39	0.00	0.00	2.34	2.21	0.00	0.00
2016	MSPD+SOMA	1.53	3.80	0.74	-0.02	0.00	0.00	2.23	2.24	0.00	0.00
2017	MSPD+SOMA	2.04	3.80	0.74	0.11	0.00	0.11	2.15	2.25	0.23	0.76
2018	MSPD+SOMA	2.73	3.90	0.74	0.21	0.00	0.21	2.04	2.30	0.43	1.49
2019	MSPD+SOMA	2.14	3.90	0.75	0.11	0.00	0.11	2.01	2.28	0.23	0.80
2020	MSPD+SOMA	0.73	3.80	0.92	-0.09	0.00	0.00	2.46	2.55	0.00	0.00
2021	MSPD+SOMA	1.00	3.80	0.97	0.03	0.00	0.03	2.46	2.14	0.06	0.19
2022	MSPD+SOMA	2.91	3.80	0.92	0.33	0.00	0.33	2.34	2.08	0.77	2.39
2023	MSPD+SOMA	4.37	3.80	0.94	0.47	0.00	0.47	2.38	2.26	1.12	3.42
2024	MSPD+SOMA	4.43	3.80	0.95	0.48	0.00	0.48	2.25	2.23	1.08	3.41
2025	MSPD+SOMA	4.20	3.80	0.95	0.46	0.00	0.46	2.18	2.22	1.01	3.27

r is the stock-structure marginal rate and g expected nominal growth, both in percent; b is consolidated interest-bearing debt/GDP. The ratios are the inflation-layer ratios $[P]/[E]$; "Treasury" covers the Treasury's own marketable debt (2001 on). 1980–2002 use Treasury Bulletin FD-5 maturity buckets with the bias corrections of Appendix A.